

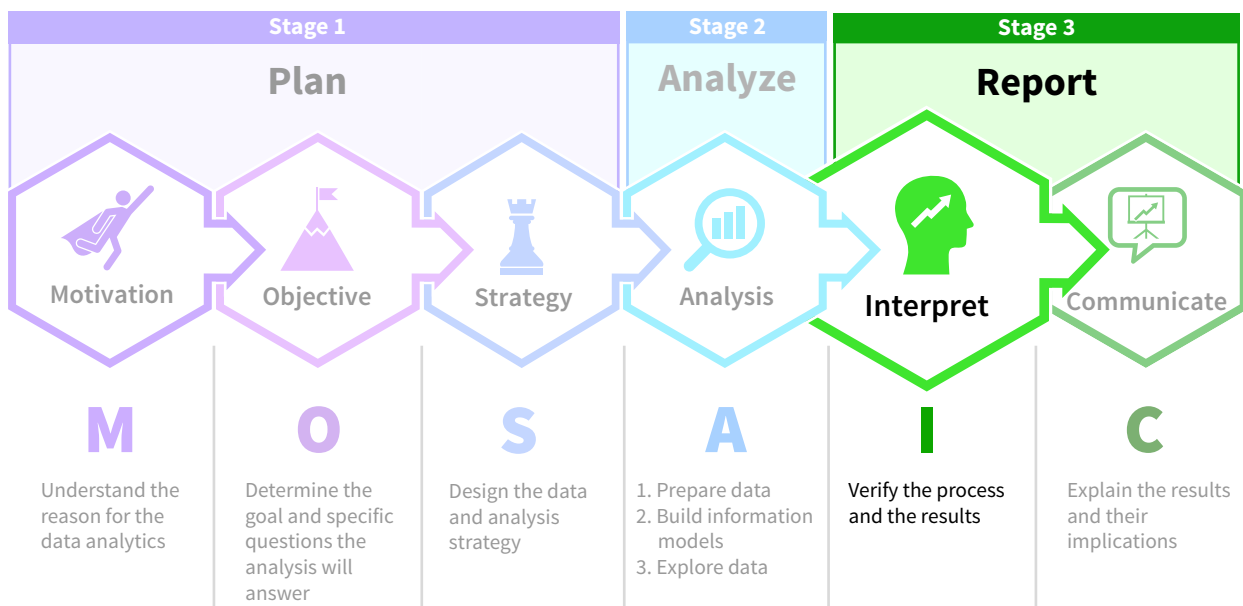
Interpreting Data Analysis Results

Chapter Preview

Your data analysis skill-building journey continues in this chapter with a focus on interpreting data analysis results. You have been developing your data analytics mindset, identifying and analyzing the motivation for data analytics projects, learning how to identify relevant questions, data, and concepts, and designing data analysis strategies. You also learned the skills necessary to prepare data for analysis, build information models, and explore data. Now, it is time to interpret the analyses' results so they can be communicated to stakeholders.

It might initially seem that data exploration and data interpretation are the same thing. While some aspects are similar, there are also key differences. The data exploration step focuses on analyzing rather than interpreting. Interpreting data analysis results is a critical step in the data analysis process.

There will be times during your career when you must interpret a data analysis that was prepared by someone else in your organization, clients, creditors, or supply chain partners. This chapter focuses on interpreting the meaning of both your own final analyses and results prepared by others.



Professional Insight: Do the Analysis Results Make Sense?

Megan, a senior tax accountant, works at one of the ten largest U.S. accounting firms.

When I first graduated, the firm I worked in outsourced all tax return and analysis preparation. Right away I was expected to review analyses prepared by other people and explain those results to the partner. It was really difficult to suddenly be in a review role rather than a preparer role. I knew how to check the numbers to see if they were correct, but interpreting what the results meant was much more challenging.

As I have transitioned into a senior role, I have seen the importance of critical thinking even more. My favorite staff to work with are those who review their results and ask questions. They don't just assume that, because the numbers were input correctly, the results are correct. **You can check all the numbers to see if they are correct, but when you do the analysis, you need to be able to determine if the results make sense.**

A staff member once input all the data correctly into the tax software and prepared a tax liability analysis that showed the client's tax liability went up. However, this didn't make sense because that client had a \$100 thousand loss from one of their businesses. So, the tax liability should have gone down. When I pointed this out, the staff member responded, "I was so caught up in the details that I rationalized the results instead of asking more questions." Stepping back to evaluate if something makes sense is difficult. As a student I didn't practice doing that. I was just focused on getting the right answers and getting a high grade.

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 8.1 Compare data analysis interpretation and data exploration.	- Data Analysis Interpretation Versus Data Exploration - Data Analysis Interpretation	Interpret Revenue Visualization (Example: Financial Accounting)
LO 8.2 Apply critical thinking to data analysis interpretation.	- Stakeholders: Understand the Context - Purpose: Define the "Why" of the Analysis - Alternatives: Investigate Other Interpretations - Risks: Consider Data, Analysis, and Bias - Knowledge: What We Need to Know - Self-reflection: Think About Lessons Learned	Evaluate a Cost Analysis (Example: Auditing)
LO 8.3 Determine if the data analysis results answer the question and align with the objective of the analysis.	- Evaluate the Data and Methods - Examine the Results - Determine if More Information or Analyses are Necessary	Interpret a Refund Analysis (Example: Managerial Accounting)
LO 8.4 Evaluate the validity and reliability of descriptive and diagnostic data analysis results.	- Descriptive Analytics - Diagnostic Analytics	Interpret a Scatterplot for Outliers (Example: Auditing)
LO 8.5 Evaluate the validity and reliability of predictive and prescriptive data analysis results.	- Predictive Analytics - Prescriptive Analytics	Interpret Regression Results (Example: Managerial Accounting)

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

8.1 How Do We Draw Conclusions from Data Analysis?

LEARNING OBJECTIVE

1

Compare data analysis interpretation and data exploration.

Data analysis interpretation is the process of evaluating an analysis to understand and explain its meaning. Similar to interpreting a weather forecast when planning an outdoor party, the insights gained from data analysis interpretation help us make good business decisions. Data analysis interpretation and data exploration probably seem very similar. In fact, you are interpreting even as you are exploring the data.

Data Analysis Interpretation Versus Data Exploration

The goal of data exploration is understanding the data, whereas data interpretation involves understanding the analysis. The focus, however, is what differentiates exploration and interpretation. An example can illustrate the difference. Imagine conducting an audit of accounts receivable. The client has given the audit team access to the available accounts receivable data. An excerpt from the data is shown in **Illustration 8.1**. The first step is exploring the data to understand it. Recall that data exploration is a process with four steps:

1. Identify questions about the data.
2. Identify data relationships.
3. Explore data relationships.
4. Generate insights.

ILLUSTRATION 8.1 Accounts Receivable Database

	A	B	C	D	E	F	G	H	I
1	Client Accounts Receivable Database Excerpt								
2	CustomerID	PaperlessDate	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	Disputed	SettledDate	PaperlessBill
3	0379-NEVHP	4/5/2025	611365	1/1/2025	1/31/2025	\$ 55.94	No	1/14/2025	Paper
4	8976-AMJEO	3/2/2024	7900770	1/25/2025	2/24/2025	\$ 61.74	Yes	3/2/2025	Electronic
5	2820-XGXS	1/24/2024	9231909	7/2/2025	8/1/2025	\$ 65.88	No	7/7/2025	Electronic
6	9322-YCTQ	4/5/2024	9888306	2/9/2025	3/11/2025	\$ 105.92	No	3/16/2025	Electronic
7	6627-ELFBK	11/25/2024	15752855	10/24/2024	11/23/2024	\$ 72.27	Yes	11/27/2024	Paper
8	5148-SYKLB	8/27/2025	18104516	1/25/2024	2/24/2024	\$ 94.00	Yes	2/20/2024	Paper
2466	7050-KQLDO	9/28/2024	9989225541	4/26/2024	5/26/2024	\$ 53.16	No	5/17/2024	Paper
2467	9758-AIEIK	4/22/2024	9990243864	7/3/2025	8/2/2025	\$ 68.66	No	7/17/2025	Electronic
2468									

The data could be explored to calculate the total accounts receivable by aging category, and the data and payment information could be used to identify relationships in the data:

- Column G represents accounts where the customer disagrees with the invoice amount.
- Exploration: Is there a relationship between accounts receivable being disputed and how overdue the payment is?

An alternative relationship could also be explored:

- Column I data show whether a customer receives an electronic or paper bill.
- Exploration: Does paper or electronic billing have a relationship with how long it takes a customer to pay?

The insight gained from analyzing the relationships can be used to estimate uncollectible accounts. Performing many exploratory analyses will eventually lead to the analysis that accurately explains the accounts receivable and uncollectible accounts.

Now, it is time to move into interpretation mode. At this point, we understand the data, but we must understand what the analyses mean in terms of supporting the audit tests of completeness and reasonableness of the allowance for doubtful accounts. Data analysis interpretation happens at the end of the data exploration process. We move from exploring for insights to interpreting so we can make informed decisions.

Data Analysis Interpretation

Data analysis interpretation is the process of reviewing an analysis. There are two steps, each with specific questions that should be answered.

Step 1: Determine if the analysis makes sense.

- Question 1: Does the analysis answer the intended question and align with the original objective?
- Question 2: Were the correct data and methods used to perform the analysis?
- Question 3: Are the results reasonable, or are more analyses necessary?

Step 2: Verify that the results are valid and reliable.

- Question 4: Does the analysis measure what it was intended to measure?
- Question 5: Are the results accurate?

These steps probably seem familiar because they mirror the data analysis process in previous chapters. When interpreting our own analysis, the first two questions in Step 1 are answered during the planning and analyzing stages of the MOSAIC process:

- Does the analysis answer the original question or objective of the analysis? (**Motivation and Objective**)
- Were the correct data and appropriate method used to perform the analysis? (**Strategy and Analysis**)

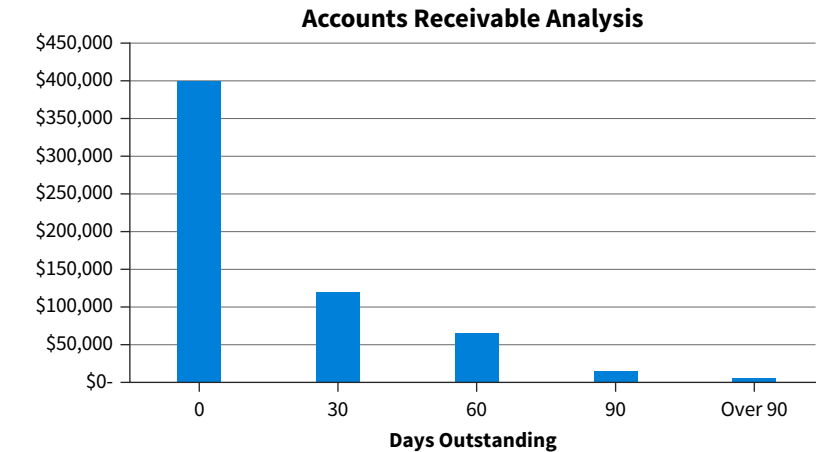
The remaining questions from both steps would then be answered during the reporting stage of the data analysis process (**Interpretation**). However, if someone else prepared the analysis, then it is necessary to address each question for both steps. This chapter works through the five questions assuming the analysis was prepared by someone else.

Let's continue with the accounts receivable example. During an audit, the audit team will review the accounts receivable balance to confirm two things:

1. The receivables exist and are legitimate.
2. The amount reflected in the balance sheet contains all the accounts receivable and is complete.

Understanding accounts receivable involves reviewing an analysis prepared by a client, which must be interpreted. **Illustration 8.2**, a column chart, is an example of an accounts receivable analysis called an accounts receivable aging report. The chart breaks total accounts receivable into categories based on when the receivables are due. Each bar in the chart represents the total of accounts receivable due for each category of due dates.

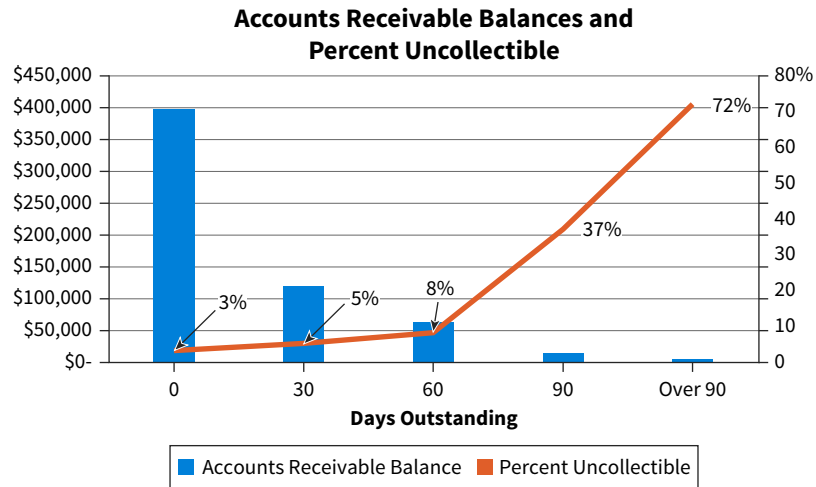
ILLUSTRATION 8.2 Accounts Receivable Aging Analysis



This analysis could be one of many that the audit team must interpret. It helps auditors understand the relationship between the accounts receivable total and the breakdown of how much of that total is past due and for how long.

Imagine the audit team is testing whether the valuation of the allowance for doubtful accounts is appropriate. The client has provided the visualization in **Illustration 8.3**.

ILLUSTRATION 8.3 Accounts Receivable Aging and Percent of Uncollectible Accounts



Some important points about this visualization:

- This is a dual axis chart, which helps illustrate the relationship between variables with different measurement scales.
- This example examines the total amount of accounts receivable in each of the five aging categories and the percentage of those receivables that are uncollectible.
- The left axis shows the amount of accounts receivables in U.S. dollars. The right axis is the percentage of receivables that are uncollectible.
- Including both variables in the visualization shows the total accounts receivable by aging category and what percent of those receivables are considered uncollectible.

Recall from your first financial accounting course that the allowance for doubtful accounts is an estimate of uncollectible accounts receivable. This amount is then deducted from the accounts receivable balance reported on the company's balance sheet.

The audit team asked the client for information related to the age of the current receivable balances (accounts receivable aging report) and an estimate of what percentage of the balances are uncollectible. Now it is time to interpret the analysis the client has provided.

One of the most valued skills of accountants is their ability to be independent and skeptical evaluators of financial information. **Illustration 8.4** is a brief example of data analysis interpretation of the visual in Illustration 8.3. Each step is covered in more detail throughout the chapter.

ILLUSTRATION 8.4 Overview of Data Analysis Interpretation

Interpretation Process	Example: Interpretation of Illustration 8.23	Learning Objective
Step 1: Determine if the analysis makes sense.		
1. Does the analysis answer the intended question and align with the original objective?	- The question is whether the allowance for doubtful accounts seems reasonable. The analysis does reveal some insight, but more information is needed. - For example, we do not know the balance in the allowance for doubtful accounts.	LO 3
2. Were the correct data and methods used to perform the analysis?	Data: - To make this judgment, check the data to confirm the correct accounts receivable data were used in the analysis. - Confirm that the data used to calculate the uncollectible percentage amounts are correct.	LO 3
	Methods: - The analysis is descriptive in nature. The totals for accounts receivable by due date are provided in the left axis, and the percent uncollectible is illustrated with a call out for each percentage by category. Determine if this is the most appropriate method. - Since the goal is determining if the valuation is reasonable, perhaps a comparison of the allowance for doubtful accounts over the past five years to account receivable balances would be more appropriate.	LO 3
3. Are the results reasonable, or is more analysis necessary?	- To make this judgement, compare the information in the analysis with what is known about the client's accounts receivable. If the totals agree with the financial statements and estimates of uncollectible accounts, then the analysis is reasonable. - We can also evaluate if the visualization is reasonable. For example, most receivables should be in the under 30-day category, and the receivable most likely to not be collectable would be in the 90 days late or over category.	LO 3
Step 2: Verify the results are valid and reliable.		
4. Does the analysis measure what was intended?	Validity refers to how well the analysis represents reality: In this example, determine if the visualization represents an analysis of the allowance for doubtful accounts valuation.	LO 4 and LO 5
5. Are the results accurate?	Accuracy means that the measures used in the analysis are correct and without mistakes: - The measures used in the visualization are account balances, days outstanding, and a calculation of the percent of uncollectible accounts. These are all correct measures. - Confirm the amounts are correct by comparing the accounts receivable to the amounts in the financial statements.	LO 4 and LO 5

Apply It 8.1
 Interpret Revenue
 Visualization



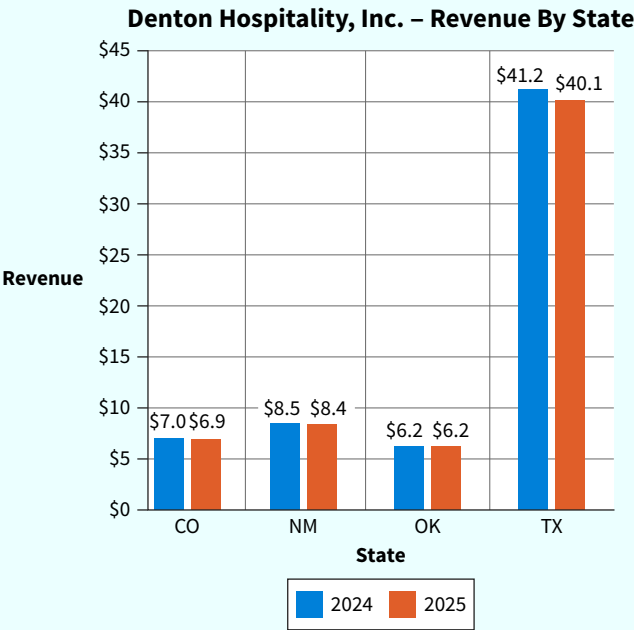
Data **Financial Accounting** Denton Hospitality, Inc. (DHI) is a hotel chain in the southwestern United States. Dante Garcia started the hotel chain in Denton, Texas in 1968 with one hotel. Since then, the chain has grown to 48 hotels in 4 states. Dante retired in 2002 and was succeeded as CEO by his daughter, Luciana.

DHI’s hotels are in the economy lodging segment. Properties are in Colorado, Oklahoma, New Mexico, and Texas. A typical economy lodging hotel has an average of 84 rooms, although DHI’s hotels average 117 rooms. Properties are staffed by a general manager, front desk staff of 6 persons, a head housekeeper, 7 housekeepers, and a maintenance worker. Except for the general manager, employees are paid hourly and their assigned hours vary based on demand. A summary of DHI’s financial performance is provided.

Summary of DHI Financial Performance

State	Number of Hotels	Revenue 2024	Revenue 2025	Profit 2024	Profit 2025
CO	5	\$ 6,972,316	\$ 6,942,074	\$ 3,701,428	\$ 3,582,689
NM	7	\$ 8,491,172	\$ 8,414,229	\$ 4,025,218	\$ 3,862,043
OK	5	\$ 6,223,433	\$ 6,156,051	\$ 3,178,742	\$ 2,928,290
TX	31	\$ 41,220,622	\$ 40,071,922	\$ 18,320,989	\$ 16,548,176
Grand Total	48	\$ 62,907,543	\$ 61,584,276	\$ 29,226,377	\$ 26,921,198

DHI would like to know how revenue compares by state and if revenues are rising or falling. Luciana has asked you to interpret this analysis.



- List the questions you would ask to interpret this analysis, and provide the answers.
- Is this exploratory data analysis or data analysis interpretation?

SOLUTION

- Does the analysis answer the question and align with the objective? Yes:
 - The question is how the states’ revenues compare and if revenues are increasing.
 - The chart shows revenue by state and a comparison to the prior year.

Were the correct data and methods used to perform the analysis? Yes:

- DHI is interested in revenue performance. This chart includes revenue.
- DHI also wants to know performance by state and if revenue is increasing. This bar chart shows that information by grouping revenue by state and showing current year compared to prior year.

Are the results reasonable? Maybe:

- It is necessary to confirm an understanding of DHI and their revenue to make a final decision about reasonableness.
- The bar chart shows how many hotels are in each state. It seems reasonable that Texas has the highest amount of revenue because about 65% of all the hotels are in Texas.

Is the analysis valid? That is, did it measure what it was intended to measure? Maybe:

- The results seem to be valid because the analysis measured what was intended—revenue by state and comparisons to prior year.

Is the analysis accurate (reliable)? Maybe:

- More research is necessary to compare the results to the financial statements to ensure all the revenue is represented and that the analysis is accurate.

2. Since DHI asked for an interpretation of the analysis, it would seem logical that this is an interpretation. If the task was interpreting this analysis to determine additional analyses that could be performed based on the outcome, it could be considered exploratory.

8.2 What is the Relationship Between Critical Thinking and Data Analysis Interpretation?

LEARNING OBJECTIVE 2

Apply critical thinking to data analysis interpretation.

In the chapter opener, Megan discussed the importance of critical thinking in data analysis. Thinking deeply about the analysis helps interpret the results and avoids being “... *so caught up in the details that I rationalized the results instead of asking more questions.*” Applying a critical thinking framework to the analysis being interpreted ensures that we are consistently thinking critically about every analysis.

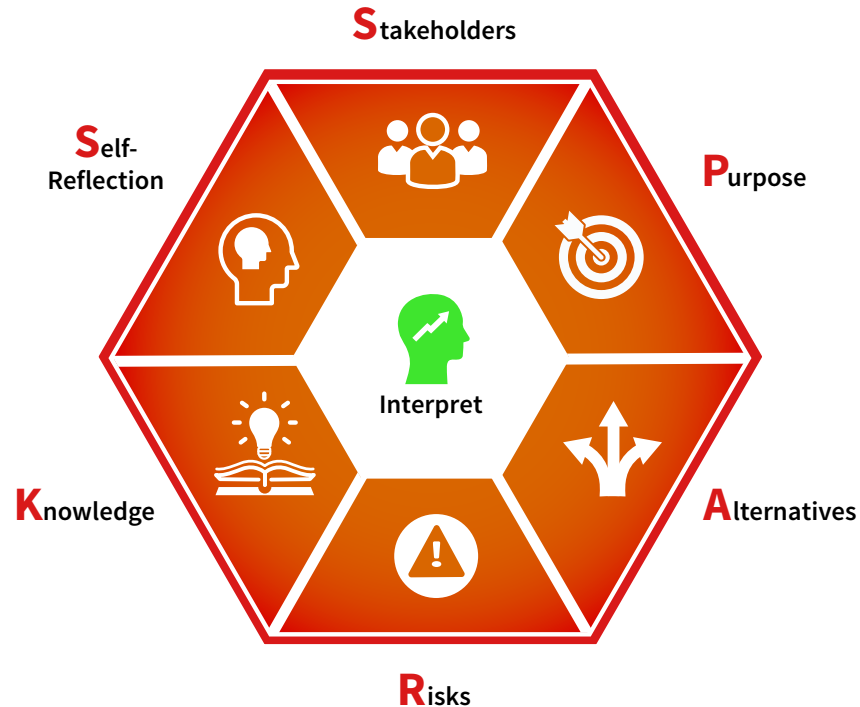
Let’s use critical thinking to interpret a data analysis prepared by someone else. The manufacturing firm Super Scooters makes four models of scooters:

- The Captain and Lazer models are electric scooters.
- The Kicks model is not powered.
- The Celeritas is a gas-powered scooter.



Super Scooters has benefited from the increase in scooter-sharing systems in large cities. Like bike-sharing, a scooter-sharing system is a service that provides scooters for short-term rentals. The global market for electric scooters is estimated to reach \$34.7 billion by 2028.¹ Companies such as Lime and Bird are purchasing large quantities of electric scooters as the market for scooter-sharing expands. Super Scooters believes the time may be right to move to manufacturing only electric scooters to potentially increase sales to companies like Lime and Bird. Imagine you are a staff accountant who is part of a team evaluating whether to continue producing the Celeritas model gas scooter. Let's apply the elements of critical thinking (**Illustration 8.5**) to this example.

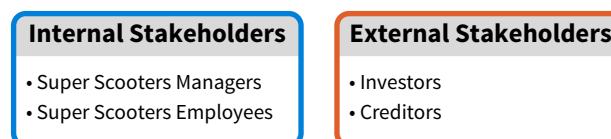
ILLUSTRATION 8.5 The SPARKS Critical Thinking Framework



Stakeholders: Understand the Context

Before interpreting the analysis provided by the Celeritas marketing team, consider who the stakeholders are in this decision. Identifying stakeholders (**Illustration 8.6**) provides insight into the situation within which the data analysis was created. This knowledge can help explain the results.

ILLUSTRATION 8.6 Super Scooters' Internal and External Stakeholders



¹<https://www.grandviewresearch.com/industry-analysis/electric-scooters-market>

Failing to identify the stakeholders means potentially interpreting the results from the wrong perspective. In this example, if you were not thinking about Super Scooters' creditors, you may not consider the impact of dropping the Celeritas scooter on the company's ability to pay their existing loans.

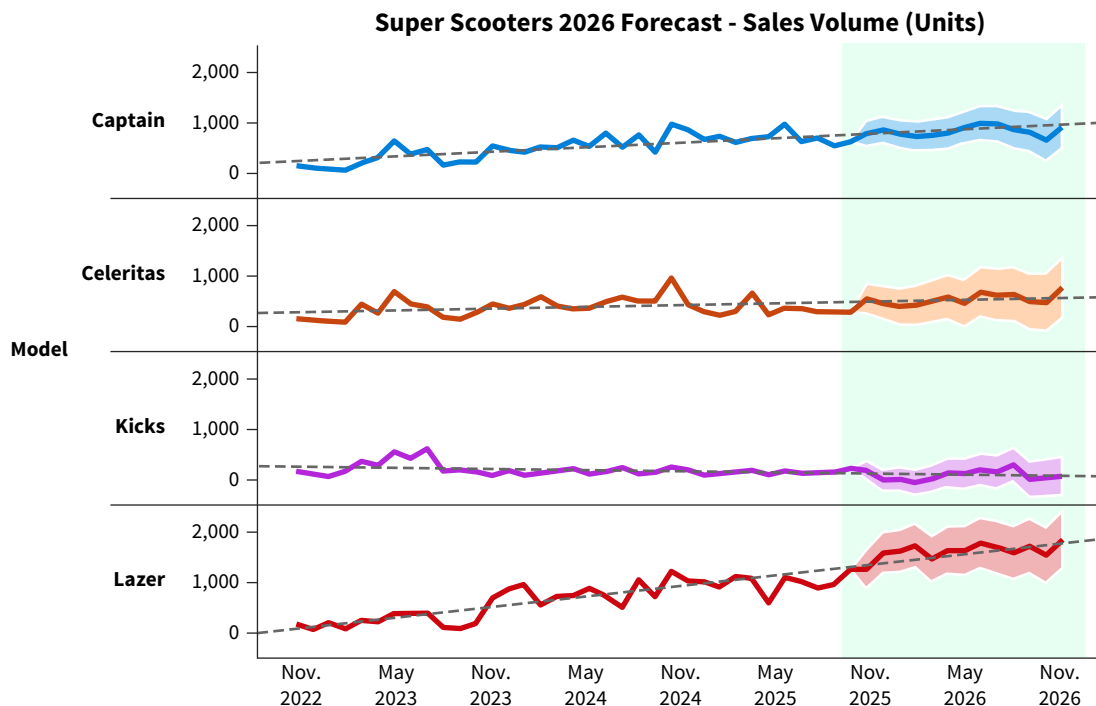
Purpose: Define the “Why” of the Analysis

In addition to recognizing relevant stakeholders, identify the purpose of the analysis. It is easy to forget this and immediately start interpreting. Be careful not to fall into this trap! We cannot fully interpret an analysis until we know its purpose.

In this example, Super Scooters is considering eliminating the Celeritas scooter. The Celeritas marketing team has prepared an analysis of sales for the last four years and a forecast for 2026. The shaded part of the line represents the 2026 forecast. The purpose of the analysis is to understand the sales trends for the four models of powered scooters that Super Scooters currently manufactures and sells.

Overlooking the purpose could mean interpreting the analysis incorrectly. If you did not know the purpose of the analysis in [Illustration 8.7](#) was to evaluate the Celeritas model, you might waste time evaluating a different model.

ILLUSTRATION 8.7 Super Scooters Forecast



Alternatives: Investigate Other Interpretations

Always consider if there are alternative ways to view the results of an analysis. Additionally, consider if there are alternative methods to conduct the analysis that were not addressed. Finally, decide if more analysis is needed. Thinking about these different alternatives helps determine whether the most appropriate analysis was used, which can increase confidence when interpreting the results.

In the Super Scooters example, alternative interpretations may include:

- The trend analysis assumptions do not include potential impact to sales of the other models should the Celeritas be eliminated.
- The trend analysis assumptions may be too aggressive.
- The trend analysis alone is not enough information on which to base a decision to keep or drop the Celeritas scooter.
- The Kicks model has the poorest performance. Perhaps that model should be evaluated for elimination instead of the Celeritas.

Risks: Consider Data, Analysis, and Bias

Examine all aspects of the analysis to identify potential risks. This begins with the data and extends to potential biases—both our own and those of stakeholders. Asking certain questions can help evaluate these potential risks. [Illustration 8.8](#) lists the risks, questions that could be asked to evaluate them, and applies a Super Scooters example. Keep in mind that if you are the preparer of the analysis, then you will have already addressed some of these issues.

ILLUSTRATION 8.8 Questions that Help Identify Risks

Potential Risks	Questions	Super Scooters Examples
Data	• Completeness: Is the analysis missing relevant data? • Accuracy: Are the data used in the analysis correct? • Timeliness: Are the data used in the analysis the most recent available? • Internal controls: Were appropriate internal controls in place to ensure the data used were correct?	• Some of the sales data were not included or may be incorrect. • The data starts in 2022 and continues through 2026, so they appear timely.
Analysis	• Method: Was the correct method used to perform the analysis? • Data: Did the analysis use the right data? • Purpose: Did the analysis answer the question?	• The purpose of the analysis was to understand sales trends for the Super Scooter models. • The analysis uses sales amounts from 2022 to 2025, so it does show how sales are trending. • The risk of using the wrong estimates to forecast the 2026 figures should be considered.
Bias	• Data biases: Were all the necessary and appropriate data included in the analysis? • Preparer biases: Did the preparer have any potential biases that could have influenced the preparation of the analysis? • Evaluator biases: Does the evaluator (you) have any biases that could influence the interpretation of the results?	• The data could be biased if the analysis did not include all the available sales data. • The preparer could be biased to provide a positive analysis if they do not want to eliminate the Celeritas model. • The evaluator could be biased in favor of eliminating the Celeritas or in favor of keeping the Celeritas model.

Knowledge: What We Need to Know

To interpret any analysis, determine the knowledge necessary to understand it. We may not have the correct background or experience, so identifying the required understanding will reveal if we need to do additional research. It helps to ask these questions:

- Is specific accounting knowledge necessary?
- Would industry knowledge be helpful?
- Is technology knowledge important to interpret the analysis?
- Is additional research necessary, or should we seek the help of an expert?

What types of knowledge are required for interpreting the Celeritas marketing team's analysis?

- **Accounting knowledge:** Understanding the sales forecast requires knowing the sales revenue and how the forecast was calculated. To determine whether to discontinue the Celeritas model, an understanding of costs, cost volume profit analyses, and how to perform a keep or drop analysis are necessary.
- **Industry knowledge:** Super Scooters operates in the automotive and transportation industry, so you would need information about the manufacturing industry of gas-powered scooters and electric-powered scooters. Learning about the supply chain for the manufacture of gas and electric scooters, as well as the competitors in scooter manufacturing, would also be important.
- **Technology knowledge:** In this example, the visualization shows the sales and sales forecast for the Celeritas model (Illustration 8.7). Understanding how this visualization was created would be helpful since that software knowledge will help you better evaluate the visualization.
- **Other knowledge:** If there is accounting, industry, or technical knowledge you need, do additional research and/or solicit the help of someone with that knowledge.

Self-Reflection: Think About Lessons Learned

Each data analysis project should include reflecting on lessons learned from previous analyses performed or interpreted. We can also learn from the current data analysis being performed or interpreted and apply that to future data analysis projects. Reflecting on previous experiences helps us perform interpretations quickly, thoroughly, and accurately.

In the Super Scooters example, reflect on other analyses you have performed that may be similar. Did you do any keep or drop analyses in your college courses? Have you analyzed sales trend analyses prior to this? If so, how can you apply those experiences to this interpretation?

Auditing Denton Hospitality, Inc. (DHI) CEO Luciana Garcia is concerned about growing costs in some of the hotels and has asked the internal audit department to evaluate which hotels have increasing costs over the past two years. As an internal auditor for DHI, your manager asked you to prepare and interpret an analysis of costs by hotel for the past two years and identify the locations with the highest increases.

List each critical thinking element and identify how you would apply it to this example.

SOLUTION

Critical Thinking Element Example



Stakeholders

Internal: Purchasing department, hotel management
External: Creditors, vendors

Apply It 8.2 Evaluate a Cost Analysis



	Purpose	Evaluate expenses by hotel site and determine which locations have experienced increases.
	Alternatives	There may be alternative analyses that could or should be prepared. There could also be different explanations for why expenses have increased. For example, increases in hotel occupancy could drive increases in hotel expenses.
	Risks	Data may be missing or incorrect. An analysis could be incorrectly interpreted.
	Knowledge	Knowledge about the company and the industry, hotel expenses, and the preparation and interpretation of data analyses is necessary.
	Self-reflection	There could be expense analyses from prior projects that could be applied to this one. Consider whether you could apply what was learned during this project to future projects.

8.3 How Do We Know the Analysis Makes Sense?

LEARNING OBJECTIVE 3

Determine if the data analysis results answer the question and align with the objective of the analysis.

Recall that the first step in data interpretation is determining if the analysis makes sense. This may seem like an obvious question, but it is one that is often overlooked. It is even more important when trying to understand an analysis of something unfamiliar. In terms of data analysis, asking if the analysis “makes sense” means confirming the analysis has a clear meaning.

Using critical thinking skills, determine if the analysis answers the intended question and aligns with objective, if the correct data and methods were used, and if the results are both reasonable and sufficient for the project’s purpose.

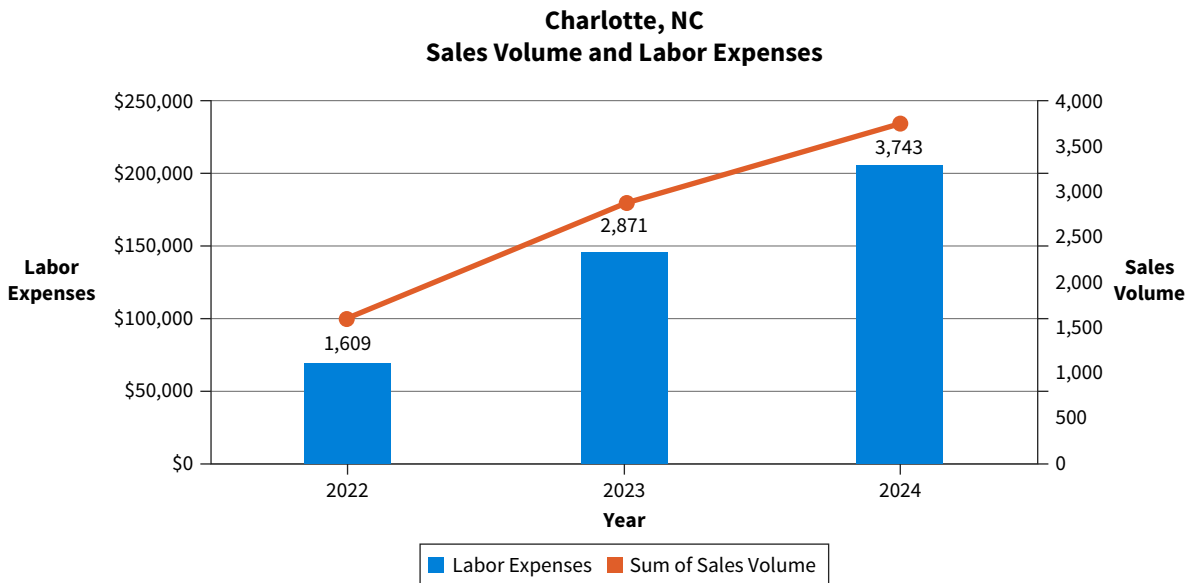
Evaluate the Data and Methods

Questions we can ask to determine if the analysis makes sense include thinking about *how* we got the results:

- Are the data used in the analysis reasonable given the question/objective of the analysis?
- Is the analysis method reasonable given the question/objective of the analysis?

Illustration 8.9 is an analysis designed to show if Super Scooters' labor expenses in the Charlotte, NC location increased as sales volume increased.

ILLUSTRATION 8.9 Dual Axis Chart of Sales Volume and Labor Expenses



Are the data used in the analysis reasonable given the question/objective of the analysis?

- The data used in this chart are labor expenses and sales volume. It is reasonable to use these two measures since we want to know if they are related. In other words, do both measures move in the same direction?

Is the analysis method reasonable given the question/objective of the analysis?

- A dual axis chart compares two different measures in one graph. Labor expense numbers (\$50 thousand–\$250 thousand) are much larger than sales volume numbers (1,000–4,000). If a clustered column chart (**Illustration 8.10**) was used instead, the sales volume bar would be too small to distinguish a relationship.

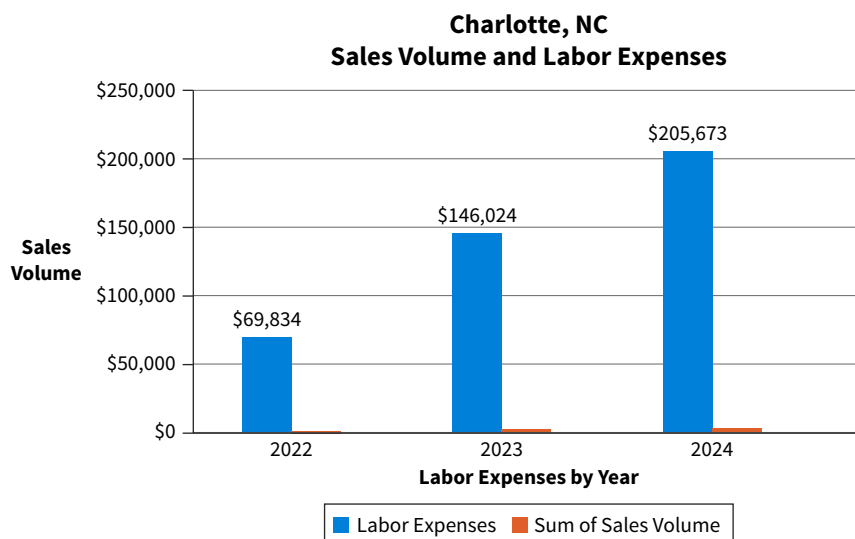


ILLUSTRATION 8.10 Clustered Column Chart of Sales Volume and Labor Expenses

Examine the Results

There are also questions that evaluate the reasonableness of the results themselves:

- Are the results of the analysis reasonable given what is known about the subject being analyzed?
 - Super Scooters manufactures the scooters they sell. It makes sense that as the number of scooters sold increases, they would need to produce more scooters. Increased production would lead to increased labor.
- Are the implications of the analysis reasonable given what is known about the subject being analyzed?
 - The implication is that if Super Scooters continue to increase sales, then they will have increased labor costs.
- Does the analysis address the needs/concerns of the stakeholders?
 - The overall concern is whether increases in sales and labor costs at the Charlotte location are related. The stakeholders include management of Super Scooters and the employees in the Charlotte locations. This analysis will help them better understand the relationship between sales volume and labor costs.

If the answers to any of these questions is “no,” then it is likely that either more or a different analysis is necessary before the results can be interpreted.

APPLYING CRITICAL THINKING 8.1: Ask if the Analysis Makes Sense

Notice the relationship between these questions and the previous discussion about applying critical thinking. To answer these questions, you need to know:

- Who is impacted by the analysis (**Stakeholders**).
- The reason for performing the analysis (**Purpose**).
- Whether the results are reasonable given what you know (**Knowledge**).
- If you can apply past experience or knowledge to the current context and whether what you learn now can be leveraged in future analyses (**Self-Reflection**).

Determine if More Information or Analyses are Necessary

Even if the interpretation makes sense, sometimes more information or additional analyses are still necessary to thoroughly answer the question. It is easy to believe we have the information necessary to decide on a course of action. This is a common human bias that Nobel Prize winning psychologist Daniel Kahneman refers to as the “what you see is all there is” bias.

This bias can be particularly strong if the data analysis supports preconceived ideas or conclusions about the question. Accountants must be skeptical evaluators of information, so we must diligently review data analyses to be sure they provide enough information and support to make a well-informed decision.

There are more biases to be aware of when interpreting data analyses:

- **Confirmation bias:** The person performing the analysis wants to prove a predetermined assumption, so they look for data that support their existing belief. The person

interpreting the analysis can also have this bias. Being aware of the preparer's potential biases, as well as our own, can help to mitigate confirmation bias.

- **Selection bias:** This bias occurs when the data used in the analysis are selected subjectively. Selection bias is a concern if the analysis being interpreted is based on a sample of data rather than the entire population. An example is when an analysis of sales transactions is based on a sample of transactions rather than on all the transactions. If the sample is not a good representation of the full population, then the results will be biased.

Continuing the Super Scooters example, additional analyses are necessary before making a keep or drop decision. Imagine that you asked for additional information about costs and were given the analysis in [Illustration 8.11](#). (Assume you have already determined that the analysis makes sense.)

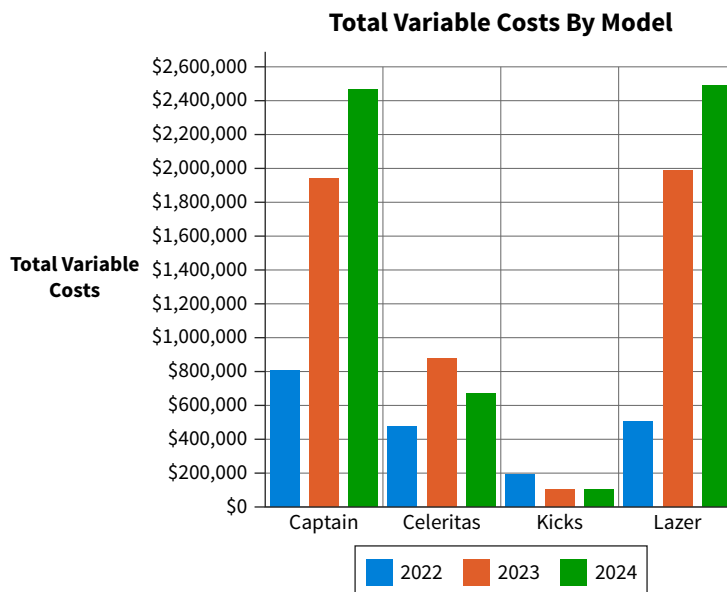


ILLUSTRATION 8.11 Variable Costs by Super Scooters Model

This analysis provides some key cost information, but do you need more? Variable costs will be eliminated if the Celeritas model is dropped, but what about fixed costs? Ask for that information as well. Do not assume that the sales and variable costs analyses are all that is needed to decide whether the Celeritas model should be dropped. There is also the potential for selection bias and confirmation bias. Consider a few questions to look for selection bias:

- Did the preparer of the sales and variable cost analyses use all the information available, or did they select just a sample of that information?
- If it was a sample, is the sample representative of all sales and all variable costs?

To mitigate the potential of confirmation bias, examine your perspective about the analyses. Was the interpretation approached with an open mind, or were you already leaning in favor of or against dropping the Celeritas model?

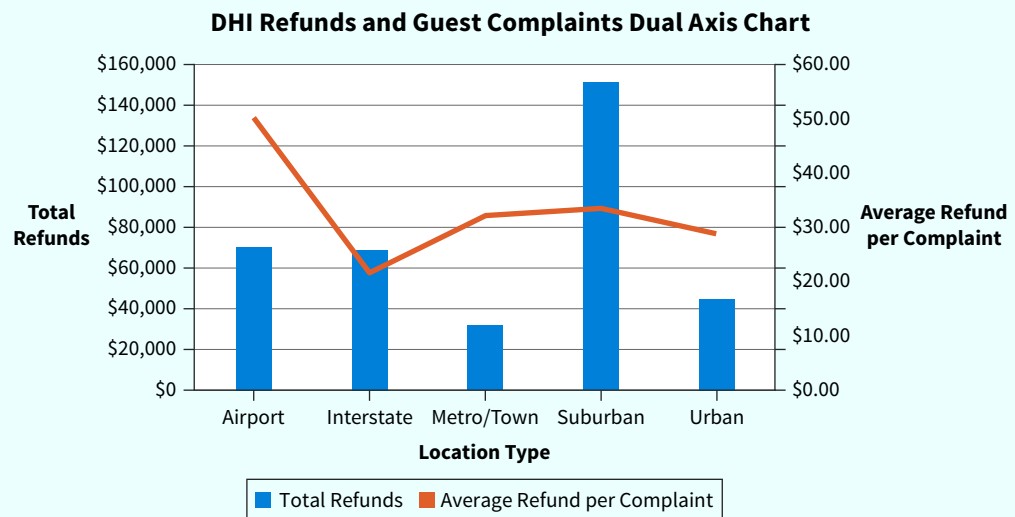
Data **Managerial Accounting** Luciana at DHI is concerned about the amount of refunds given to customers. She has asked you to speak to the hotel area managers to gain more insight, and then interpret an analysis prepared by another employee to identify the type of location with the highest amount of refunds. After speaking with the hotel area managers, you learn:

- Front desk employees can give refunds to hotel guests without approval from the general manager.

Apply It 8.3
Interpret a Refund
Analysis



- Area managers think the number of guest complaints is strongly related to the total amount of refunds. If there are many guest complaints, there will likely be a higher amount of total refunds.



1. Are the data used in the analysis reasonable?
2. Is the analysis reasonable?
3. Are the results of the analysis reasonable given what you know about the subject being analyzed?
4. Are the implications of the analysis reasonable given what you know about the subject being analyzed?
5. Does the analysis address the needs and concerns of the stakeholders?
6. Is more analysis needed? If yes, what analyses would you like to see before making a decision?

SOLUTION

1. The data seem reasonable. The purpose is understanding guest complaints and refunds. Both of those items are included in the analysis. Average refunds per complaint expresses the refund amount on a per complaint basis.
2. The objective is to identify the location type with the highest amount of refunds. The analysis does show total refunds given on one axis using bars for each location's refunds.

The analysis also shows average refund per complaint as a line with average refunds per complaint on the right axis. This is a reasonable way to evaluate the relationship between complaints and refunds because using the average refund per complaints helps look at each location based on its performance rather than size.
3. The area managers were convinced that the number of complaints is related to the total amount of refunds. The analysis shows the average amount of refunds is highest for airport locations even though the suburban locations have a higher total.
4. There are much higher total refunds in the suburban locations, so it might seem they would have the highest number of complaints. However, if there are more hotels in that category, that may also be driving the total number of complaints. By using average refund per complaint, we can see that airport locations are giving very high refunds to guests even though the total refund amount is not as high as suburban locations.
5. The analysis addresses the needs of the stakeholders to an extent. The analysis is one piece of this puzzle.
6. Yes, more analysis is necessary. To be sure that we have addressed the stakeholders, further analysis is required to address the concern about hotels offering too many refunds.

8.4 How are Validity and Reliability Determined in Descriptive and Diagnostic Analyses?

LEARNING OBJECTIVE 4

Evaluate the validity and reliability of descriptive and diagnostic data analysis results.

Once it is clear that the analysis makes sense, the results can be assessed for validity and reliability. If the results are not valid, then it does not matter how “good” the analysis is. Sometimes it is easy to be fooled by data analyses because we take them at face value. Recall the example at the start of the chapter. Megan’s staff tax accountant checked the numbers and assumed the analysis was correct. However, the analysis result was an increase in tax liability when it should have been a decrease. The staff accountant should have been more skeptical of the analysis and considered that a large loss should have caused a decrease in the tax liability.

First, what does valid and reliable mean in the context of data analysis?

- Evaluating the **validity** of a data analysis involves confirming that it measures what it is supposed to measure and represents reality.
- **Reliability** means that the data used are dependable and trustworthy and the measures used in the analysis are consistent and accurate. In this context, **accurate** means the measures used in the analysis are correct and without mistakes.

During your career you will interpret the validity and reliability of different types of analyses. **Illustration 8.12** summarizes the types of analyses most common in the four areas of analytics.

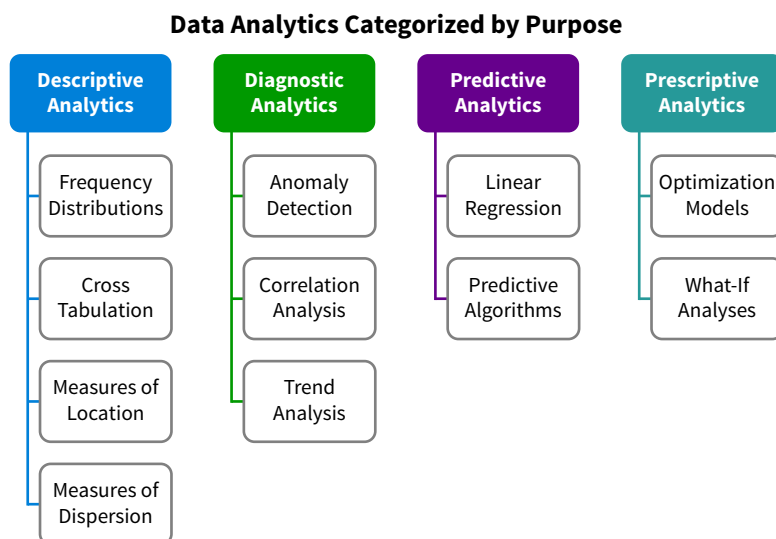


ILLUSTRATION 8.12 Summary of Analyses by Analytics Area

This section reviews the most common types of analyses used in descriptive and diagnostic analytics, how to identify appropriate measures for each type, and how to determine if the results are valid.

Descriptive Analytics

Descriptive analytics help to better understand the data underlying the analysis being interpreted. To assess validity and reliability of the analyses, ensure the correct method is used and that the data are accurate. **Illustration 8.13** is a summary of valid descriptive analyses based on the objective of the analysis. For an analysis to be valid, the method must match the objective. In a reliable analysis, the measures used are accurate and consistent.

ILLUSTRATION 8.13 Descriptive Objectives, Questions, and Valid Analyses

Objective	Example Questions	Valid Analyses
Understand categories of data.	What product category is selling the most? Which business unit has the highest revenue?	Frequency distribution Histogram
Summarize by categories and subcategories.	Which products are selling the most in each region? How much were total expenses each year by business unit?	Cross tabulation analysis
Identify an average observation in the data.	What is the average tax liability for each business unit?	Mean (if no outliers) Median
Evaluate the distribution of data.	How different are the expenses in each business unit from the company average?	Standard deviation

Understand Categories of Data

Grouping data into categories is sometimes part of an analysis. An example is when the goal is to understand sales by product. If the analysis is based on groups or categories of data, a valid analysis would be a frequency distribution or a cross tabulation analysis.

A **frequency distribution** reveals how many times something has occurred within a group or interval. The example in **Illustration 8.14** was prepared using the Super Scooters data set. (**Data** How To 8.1 at the end of the chapter shows how to create this visual in Power BI.) The frequency column reports sales orders for each model for the years 2023–2025. The relative frequency column indicates the percentage of total sales orders for each model.

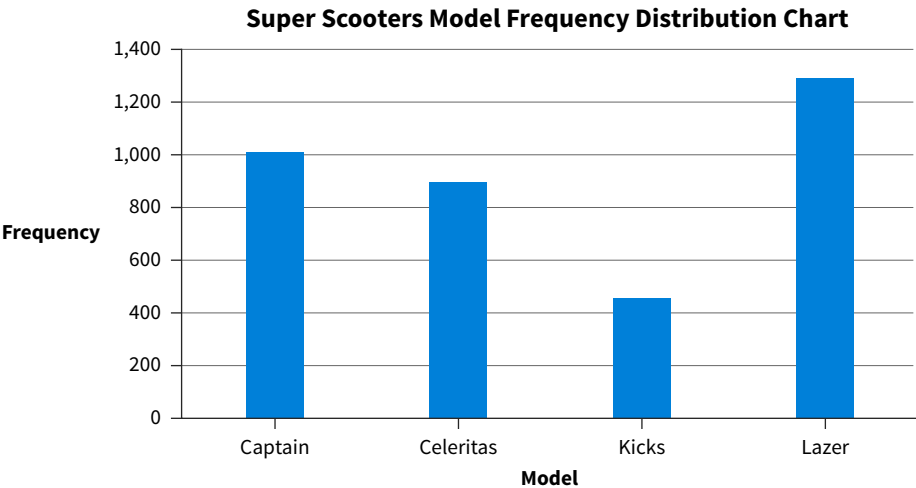


ILLUSTRATION 8.14 Super Scooters Model Frequency Distribution

Model	Frequency	Relative Frequency
Captain	1,010	27.7%
Celeritas	892	24.5%
Kicks	456	12.5%
Lazer	1,287	35.3%
Total Sales	3,645	100.0%

Sometimes frequency distribution analyses are in chart form (**Illustration 8.15**).

ILLUSTRATION 8.15 Super Scooters Model Frequency Distribution Chart



Summarize by Categories of Data

Suppose the objective is knowing the number of orders each year, from 2023 to 2025, by scooter model. A **cross tabulation analysis** would answer that question. A cross tabulation analysis shows the number of observations in a data set for different subcategories (**Illustration 8.16**).

Number of Orders Model	Year			Grand Total
	2023	2024	2025	
Captain	244	397	369	1,010
Celeritas	248	327	317	892
Kicks	261	107	88	456
Lazer	207	517	563	1,287
Grand Total	960	1,348	1,337	3,645

ILLUSTRATION 8.16 Super Scooters Cross Tabulation Analysis of Sales Orders by Model

Note that a cross tabulation analysis does not have to be in table format. **Illustration 8.17** is a column chart visualization of the table in Illustration 8.16.

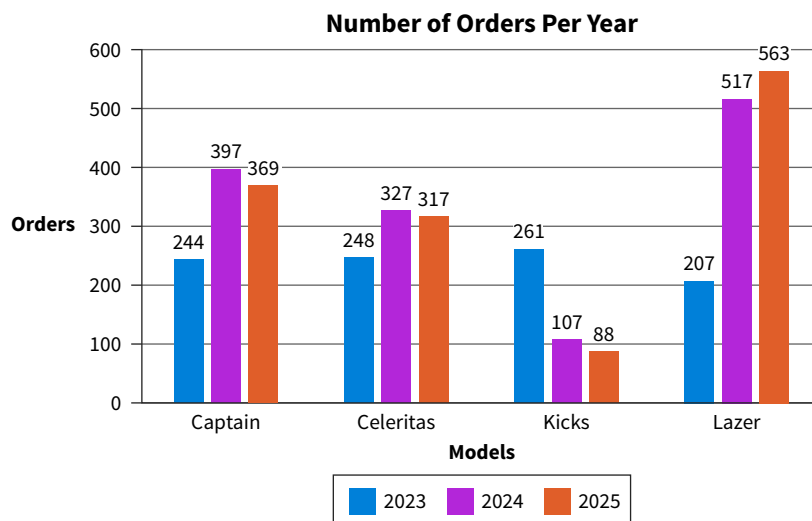


ILLUSTRATION 8.17 Super Scooters Cross Tabulation Analysis Column Chart

What can we interpret from these analyses? First, the Lazer is the best-selling model, accounting for 35.3% of total sales volume (Illustration 8.14). Second, the Lazer is the only model that shows an increase in sales volume each year from 2023 to 2025 (Illustration 8.17). Are these analyses valid and reliable? Since the objective is analyzing sales volume by model, then the frequency distribution and the cross tabulation analyses are valid methods to use. Reliability will be determined by confirming the totals agree to the actual sales volume in the financial records.

Identify an Average Observation in the Data

Sometimes the analysis question is not about groups or categories of data, but instead asks about the averages in the data. In addition to frequency and cross tabulation, measures of location are used in descriptive analytics. When evaluating the validity and reliability of an analysis that includes measures of location, ensure that the correct measure is being used.

Measures of location include the mean, median, and mode measures that reveal the average or typical observation in the data set:

- **Mean:** The sum of all observations in a data set divided by the total number of observations.
- **Median:** The middle value when the data is arranged from smallest to largest.
- **Mode:** The observation that occurs most frequently.

All three measures are valid measures of location, but under different conditions some measures of central tendency will be more appropriate than others.

For example, a mean can be affected by extreme values in the data, whereas the median is not affected by these kinds of values, which are **outliers**. There are two methods to determine if there are outliers in a data set:

- Search for a large difference between the mean and median.
- Plot the data to visually determine if there is an outlier.



Let’s start by examining the mean and the median. **Illustration 8.18** shows descriptive statistics for Super Scooters’ monthly sales volume amounts. (**Data** How To 8.2 illustrates how to create these statistics in Microsoft Excel.) The mean monthly sales volume is 2,132 and the median monthly sales volume is 1,879. The mean is much larger than the median. But we know that if we are searching for the central location of the data set, the median is right in the middle. This large difference is a good indication that there are extreme values in the data set.

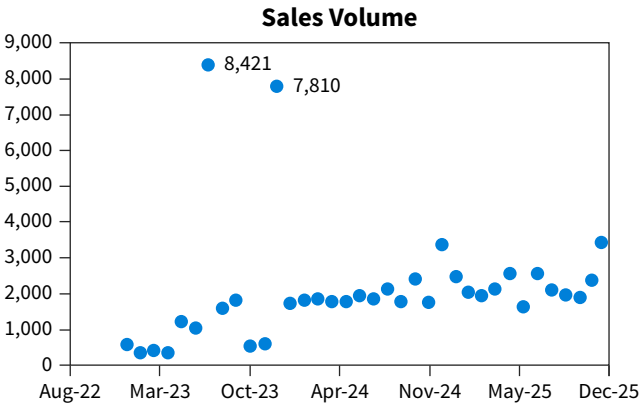
ILLUSTRATION 8.18 Descriptive Statistics for Super Scooters’ Sales Volume

Super Scooters, Inc.	
Monthly Sales Volume	
Mean	2,132.39
Standard Error	275.34
Median	1,879.00
Mode	#N/A
Standard Deviation	1,652.02
Sample Variance	2,729,183.96
Kurtosis	8.82
Skewness	2.73
Range	8,050.00
Minimum	371.00
Maximum	8,421.00
Sum	76,766.00
Count	36.00

Now try the second method. A **scatterplot** shows the relationship between two numerical variables. Each observation in the data set is plotted as a point whose coordinates relate to the values of the two variables for that observation.

Illustration 8.19 is a scatterplot of Super Scooters’ monthly sales volume from 2023 to 2025. Each sales transaction is represented by a dot. Notice the dots labeled 8,421 and 7,810. These observations are outside most of the monthly amounts. Those months are driving up the mean.

ILLUSTRATION 8.19 Scatterplot of Super Scooters’ Sales Volume



When evaluating measures of location, keep in mind the potential impact of outliers on the results. If you were responsible for creating the sales budget for Super Scooters, then you might choose between using the mean or the median sales volume from prior year to budget

for the next year's production budget. (Refer to Illustration 8.18 for the calculation of mean and median for Super Scooters' descriptive statistics.)

Illustration 8.20 summarizes the two options. Using the mean as a measure of the average amount of monthly sales production needs overstates production needs.² There is a difference of 3,036 (13%) between the two options. It would be difficult to explain to your boss next year why you were off by 13% on projected production!

Estimated Sales Based on Mean Sales Volume	Estimated Sales Based on Median Sales Volume
$2,132 \times 12 \text{ months} = 25,584$	$1,879 \times 12 \text{ months} = 22,548$
Difference Between Estimated Sales Volume = 3,036	

ILLUSTRATION 8.20 Estimated Sales Using Mean vs. Median Sales Volume

In summary, when evaluating the validity and reliability of an analysis that includes measures of location, confirm the correct measure is being used. If the correct measure is used, then the results are valid. In Illustration 8.20, using the mean instead of the median to estimate sales volume would not be valid because of the difference between mean and median monthly sales volume. Using the mean sales volume inflates the estimate due to the outliers of higher sales volume in the data. That is not a reliable budget estimate.

Evaluate the Distribution of Data

The final type of descriptive analyses discussed here are measures of dispersion. Dispersion refers to the amount of variation in the data. Are the data spread out or are they compact? In other words, how far apart are all the observations (data points) from the mean? The two most widely used measures of dispersion are variance and standard deviation.

The **standard deviation** shows how spread out the data are from the mean. It is in the same units as the mean. Standard deviation can help determine validity and reliability of an analysis. **Illustration 8.21** shows a list of descriptive statistics for Super Scooters' sales volume. Standard deviation is highlighted.

Super Scooters, Inc.	
Monthly Sales Volume	
Mean	2,132.39
Standard Error	275.34
Median	1,879.00
Mode	#N/A
Standard Deviation	1,652.02
Sample Variance	2,729,183.96
Kurtosis	8.82
Skewness	2.73
Range	8,050.00
Minimum	371.00
Maximum	8,421.00
Sum	76,766.00
Count	36.00

ILLUSTRATION 8.21 Descriptive Statistics for Super Scooters' Sales Volume

In this example, the mean amount of monthly sales volume is 2,132.39. The standard deviation is 1,652.02. So, the average distance between an observation (in this case a sale volume amount) and the mean is 1,652.02. Standard deviation can indicate the data points' relationship to the mean:

- A low standard deviation indicates the data points tend to be close to the mean.
- A high standard deviation indicates the data points are spread out over a large range of values.

²Keep in mind that this is a very simplified example of preparing a budget. Sales budgets include more than just estimating a single sale transaction amount, particularly if there are multiple products and locations. However, the danger of using the wrong measure is still a valid concern.

Does the amount of 1,652.02 seem like a high or a low standard deviation for sales volume? Considering that the standard deviation is about 77% of the mean, it does seem to be a high deviation. Review the scatterplot in Illustration 8.19 to visually determine if there are many observations (data points) far from the mean of 2,132.39. The scatterplot does show data points that are much higher than the mean.

Taken together, descriptive analytics help us better understand the data underlying the analysis being interpreted. Assessing validity and reliability of the analyses involves ensuring the correct method is used and that the data are accurate. Frequency distributions and cross tabulations can help group data into meaningful categories. Measures of location can show what the average observation looks like or if there are outliers, while measures of dispersion reveal the distribution of the data underlying the analysis.

APPLYING CRITICAL THINKING 8.2: Evaluate Reliability and Validity

- It may not seem like you are applying critical thinking skills when evaluating the reliability and validity of descriptive analytics, but you are thinking critically before you even begin:
- You must identify the reason behind the analysis to decide if it is a question that should be answered with descriptive analytics (**Purpose**).
 - Then, you identify the information necessary to interpret the analysis. Do you need to know how to interpret frequency distributions? Cross tabulations? Measures of location? Measures of dispersion? Do you have that knowledge (**Knowledge**)?
 - You could apply experiences with past interpretations of analyses to the current context (**Self-reflection**).

Diagnostic Analytics

If the objective of the analysis is to understand *why* something has occurred, then you will be interpreting diagnostic analytics. [Illustration 8.22](#) provides some common diagnostic objectives and the analyses that could be used.

ILLUSTRATION 8.22 Diagnostic Objectives, Questions and Valid Analyses

Objective	Example Questions	Valid Analyses
Find anomalies in the data set.	Are there revenue transactions that are different or suspiciously high?	Scatterplot Boxplot
Examine relationships in the data.	Is there a relationship between maintenance expense and equipment hours?	Scatterplot Correlation coefficient
Identify patterns in the data set.	Is there a seasonal sales pattern?	Bar chart Column chart Line graph Trend line

Find Anomalies

Assessing validity and reliability includes identifying any potential anomalies in the data. An **anomaly** is an observation in the data set that deviates from what is normal or expected.

Identifying anomalies is a common diagnostic analytic procedure. An anomaly may seem the same as an outlier, discussed in the section on descriptive analytics. However, there are some key differences:

- **Outlier:** This is a legitimate observation that lies an abnormal distance from other values in the data. For example, in retail stores sales are much higher during the holidays than during the summer. A high sale amount in December may be an outlier, in that it is much higher than the other months, but it is legitimate because sales are expected to be higher in December.
- **Anomaly:** This is an illegitimate observation. Illegitimate observations could be mistakes or unusual occurrences that we do not expect to see again. These observations may be outliers that are determined to be illegitimate, hence an anomaly. An example of this is if supplies expense is significantly higher in one month due to an erroneous journal entry.

Detecting anomalies could be the purpose of the analysis, such as when searching for unusually large transactions in an audit where fraud is suspected. However, detecting an anomaly could also be unintentional.

Recall that Illustration 8.19 identified two potential outliers—the sales volume amounts of 8,421 and 7,810. These amounts are much larger than the other monthly volume amounts that occurred over the 2022 to 2025 period. Why are those sale volume amounts so high? This is an example of anomaly detection. These two points might be legitimate observations (outliers), or illegitimate (anomalies).

The next step in the interpretation is to determine the reason for the high monthly volume amount and investigate other potential anomalies in the data. First, examine all the amounts that are over 1,500 and determine if they are accurate. If they are accurate, then investigate potential reasons for the high volume.

Examine Data Relationships

Part of understanding data is knowing how the data are related. **Correlation analysis** shows relationships in the data by measuring the linear relationship between two variables. If the goal of the analysis is to determine the strength of a relationship between objects of interest (variables), then a correlation analysis would be appropriate. Assessing validity and reliability of a correlation analysis requires knowing how to interpret the correlation coefficient.

Linear correlation is measured by the **correlation coefficient**, also known as the Pearson product-moment correlation coefficient. This measure is a numerical value between -1 and $+1$. The higher the absolute number, the greater the strength of the relationship. A positive correlation coefficient indicates that as one variable increases, so does the other variable. There is a positive relationship between ice cream sales and outdoor temperature, for example. As the temperature rises, ice cream sales tend to rise as well.

A negative correlation is an inverse relationship. As one variable increases, the other decreases and vice versa. Consider sales of soup. There is a negative relationship between soup sales and temperature. As temperatures decrease, the sales of soup increase. **Illustration 8.23** is a guide to interpreting the value of the correlation coefficient.

Correlation r	Interpretation
Exactly -1	A perfect negative linear relationship
-0.70	A strong negative linear relationship
-0.50	A moderate negative linear relationship
-0.30	A weak negative linear relationship
0	No linear relationship
$+0.30$	A weak positive linear relationship
$+0.50$	A moderate positive linear relationship
$+0.70$	A strong positive linear relationship
Exactly $+1$	A perfect positive linear relationship

ILLUSTRATION 8.23

Interpreting Correlation Coefficients

Correlation is a valid measure for examining linear relationships between variables in data, and the interpretation of the correlation should be based on the correlation coefficient. Correlation analysis is reliable if the correlation coefficients are consistent and accurate relative to the data being analyzed. In the ice cream and soup examples, the correlation is valid because we are examining a logical correlation using the correlation coefficient. It is reliable if we believe the correlation coefficients are consistent and accurate.

A correlation analysis can be performed using the Super Scooters data:

- Super Scooters management is reviewing variable marketing costs and would like to know if there is a relationship between the amount spent on variable marketing (discounts given as incentives for dealers to buy Super Scooters products) and sales volume.
- Management is concerned that they spent \$1.1 million on variable marketing in 2024, and they are not sure if that spending resulted in increases in sales.

Illustration 8.24 is a summary of the correlation coefficients for gross sales, sales volume, and variable marketing expenses.

The correlation coefficient for variable marketing and gross sales is 0.9650. The interpretation of that number is that there is a strong positive correlation between gross sales and variable marketing costs. As variable marketing costs increase, so do gross sales. There is also a strong correlation between sales volume and variable marketing with a correlation coefficient of 0.77410. While we cannot prove causation, we can say that there appears to be a strong positive correlation between the amount spent on variable marketing and both sales volume and gross sales.

ILLUSTRATION 8.24
Correlation Coefficients Summary
for Super Scooters

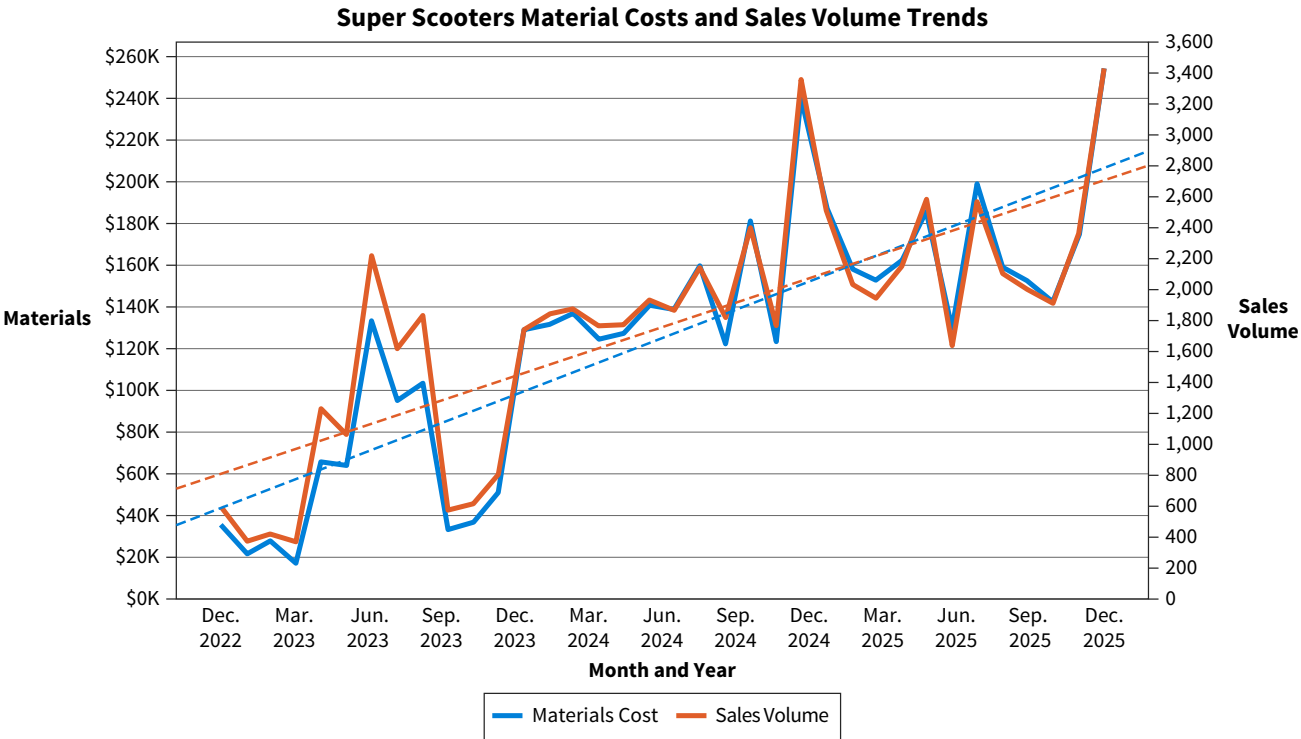
	Correlation Coefficient
Gross Sales	0.9650
Sales Volume	0.7749

Identify Patterns

A **trend analysis** is a statistical tool that uses historical data to identify patterns. It can explain why something is happening. A **trendline** indicates the general course or tendency of the data and is created using the historical data points to estimate a line.³ Examining trends helps detect patterns and relationships, which can identify opportunities or potential threats to a business.

The best way to identify trends is to graph the data over time. **Illustration 8.25** is a trend analysis of material costs and sales volume for Super Scooters for the years 2023–2025.

ILLUSTRATION 8.25 Super Scooters' Material Costs and Sales Volume Trend Analysis



³Trend analysis can also be used as a predictive analytic method.

The trend analysis was prepared with data visualization software. A trend analysis can also be prepared in Microsoft Excel using the trendline tool that is available when the data is charted.

Super Scooters is examining why material costs have been increasing. The analysis in Illustration 8.25 shows that sales volume and material costs move in the same pattern over the years and that both are increasing:

- It is logical that sales volume increases will result in rising material costs. There are similar peaks and valleys in the lines. This is an indication that there could be a seasonal pattern to sales.
- The trend lines in the analysis (the dashed straight lines) reveal that while both trend lines are increasing, material costs are increasing at a higher rate.

After reviewing this analysis, we can conclude that material costs are increasing due to increases in sales volume.

This analysis of sales volume and material costs is valid because it uses the appropriate method for understanding the relationship between trends of sales and expenses. The measures used in the analysis are accurate and consistent and the data is dependable and trustworthy, so the analysis is also reliable. However, understanding why material costs are rising at a faster rate than sales will require more investigation.

APPLYING CRITICAL THINKING 8.3: Interpret Diagnostic Analytics

When explaining why something happened, use diagnostic analytics:

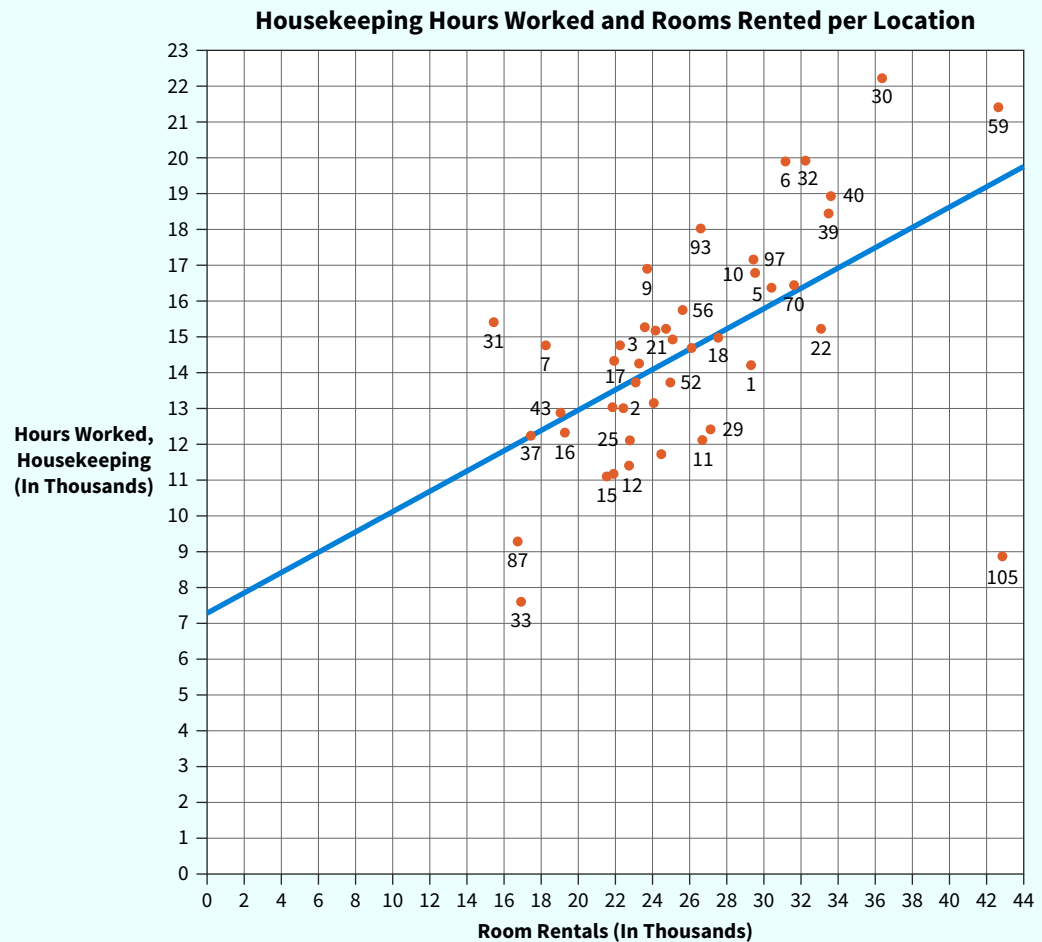
- The results of diagnostic analyses can be used to develop other explanations. For example, sales trends seem to alternate with increases and then decreases the following quarter. Perhaps sales promotions are driving this pattern (**Alternatives**).
- Look for potential threats to the analyses, such as anomalies. Are there any observations in the trend analysis that are unusual or unexpected (**Risks**)?
- Consider what is required to understand analyses, such as correlation analysis and trend analysis (**Knowledge**).

Data Auditing Roberto Jimenez is the operations manager for DHI. He has asked the internal audit department to help him perform an analysis of housekeeping hours. Roberto would like to know how efficiently different locations are performing. You have been given a scatterplot that displays the housekeeping hours worked and the number of rooms rented by hotel location. The line through the graph is a trendline indicating the linear relationship between hours worked and rooms rented. Note that the numbers listed below the dots are the hotel location numbers. All data in the scatterplot are legitimate observations.

1. Review the scatterplot, identify potential outliers, and explain why you identified them as outliers.
2. Recommend how to address the outliers.

Apply It 8.4 Interpret a Scatterplot for Outliers





SOLUTION

1. There is a positive relationship between housekeeping hours and room rentals. As the number of rooms rented increases, so does the number of housekeeping hours. This makes sense because rooms that have been rented must be cleaned.

There are several hotel location observations that are further from the trendline than other observations:

- Location 30 has the highest hours of housekeeping, but not the highest number of room rentals.
- Location 105 has low housekeeping hours but a high number of room rentals.

2. Recommendations:

- Further investigation into Location 30 to identify the cause of inefficient housekeeping hours.
- Further investigation into Location 105 to identify efficiencies that could be applied to less efficient locations.

8.5 How are Validity and Reliability Assessed in Predictive and Prescriptive Analyses?

LEARNING OBJECTIVE 5

Evaluate the validity and reliability of predictive and prescriptive data analysis results.

As you have learned, the other two types of analytics are predictive and prescriptive analytics. The next section covers the types of analyses in these areas that you are likely to encounter in your career and how to determine if the analyses are reliable and valid.

Predictive Analytics

There are many types of predictive analytics, but they all have the goal of predicting a future outcome. In the accounting profession the most common predictive analysis is **linear regression**, which is a tool for building mathematical and statistical models to explain the relationship between a dependent variable and one or more independent variables. Even if you do not prepare linear regression models in your career, understanding regression can help understand the predictive models you encounter.

Modeling Relationships

Predictive analytics build a model to help predict or better understand a phenomenon. Building a model that predicts supplies expense would help understand the factors that influence supplies expense. Building this model requires identifying the variables that will be included in it.

Once the regression model is complete, how do we know if it is valid? Remember that an analysis is valid if it measures what it is supposed to measure and if it also represents reality. Consider the variables in the model—do they make sense based on the model's goal?

For example, when evaluating a model that examines the impact of temperature on winter coat sales, it would make sense if both temperature and the average price of coats were included in the model. The model might also include snowfall. However, if the model includes a variable that does not make sense, such as number of swimsuits sold, the model will not be valid. It would not be measuring what it is intended to measure.

The next step is confirming the model is reliable. Recall that reliability means that the measures used in the analysis are accurate and consistent and the data are dependable and trustworthy. Fortunately, there are many statistical measures in a regression analysis that can be examined to determine reliability. Previous chapters described how to build regression models and interpret regression results. Here, we focus on the key statistics and output that help evaluate and interpret regression model output. Keep in mind that if the model is not valid, then it does not matter if it is reliable. An accurate and consistent measure of the model does not mean the model represents reality. So, this first step of determining if the model variables are logical is important.

Reliability of the Regression Model

Assess the reliability of regression analysis by reviewing the model's statistics. **Illustration 8.26** is the output of a multiple regression model to predict purchasing department expenses for Super Scooters. Note that purchasing department expenses are those expenses incurred

by the purchasing department to process purchase orders (payroll, administrative, and overhead). These are not the same as the cost of the purchase. The model is based on historical data, collected from each production location, for variables believed to influence total purchasing department expenses.

ILLUSTRATION 8.26 Super Scooters Purchasing Department Expense Regression Model

Super Scooters Purchasing Department Expense Regression

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.892897453				
R Square	0.797265861				
Adjusted R Square	0.777957848				
Standard Error	1,337.156824				
Observations	24				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	147659116.6	73829558.29	41.29196782	5.2812E-08
Residual	21	37547755.8	1787988.371		
Total	23	185206872.4			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	-994.5719771	877.1183137	-1.133908575	0.269611889	
# of Purchase Orders	180.0127037	31.32217805	5.747132382	1.05165E-05	
Sales Volume	1.191401172	0.351035522	3.393961854	0.002736568	

In this model, total purchasing department expense is the dependent variable, and sales volume and the number of purchase orders processed are the independent variables.

The regression in Illustration 8.26 was performed using Microsoft Excel. The Summary Output is split into three sections. First are the **regression statistics**, which are the statistical measures used to evaluate the model. **Illustration 8.27** shows the regression statistics from Illustration 8.26.

ILLUSTRATION 8.27 Regression Statistics for Purchasing Dept. Expenses

Regression Statistics	
Multiple R	0.892897453
R Square	0.797265861
Adjusted R Square	0.777957848
Standard Error	1,337.156824
Observations	24

All the regression statistics provide insight into a regression model. Each of these statistics were covered in the chapter devoted to foundational data analysis skills. Here are some of the most important statistics to evaluate for model reliability:

- **Adjusted R square (R^2):** Explains how well the regression line fits the data. The adjusted R^2 is a statistic that modifies the value of R^2 by incorporating the sample size and the number of independent variables. In general, use adjusted R^2 to evaluate a multiple regression model. The closer the R^2 is to 1, the better the fit of the regression line to the data.
- **Standard error:** In Excel regression output, standard error represents the variability of the observed dependent variable values from the values that are predicted by the model. In other words, it compares the actual dependent variable to the predicted value that the model provides. If the data are clustered close to the regression line, then the standard error will be small. If the data are more scattered, then the standard error will be larger. A small standard error is optimal.

Illustration 8.27 shows the regression statistics for Illustration 8.26:

- The adjusted R square in Illustration 8.27 is 0.778. We would interpret that 77.8% of total expenses can be explained by how many purchase orders are processed and by sales volume.
- The standard error in this model is \$1,337.16. To determine if this is a large or a small standard error, compare it to the standard deviation of the dependent variable. In this example, compare the standard error to the standard deviation of total expenses. **Illustration 8.28** provides the mean and standard deviation for total expenses.

Super Scooters Purchasing Department Costs 2022 – 2024	
Mean	\$ 3,725.30
Standard Deviation	\$ 2,837.69

ILLUSTRATION 8.28 Super Scooters Purchasing Department Expenses

- The standard deviation of \$2,837.69 is higher than the standard error of \$1,337.16 in the regression model. The standard error in this model would be considered somewhat small.

The next section of regression summary output is the analysis of variance (ANOVA) output. **Illustration 8.29** is the ANOVA section from the regression model. The ANOVA is a test for the significance of the entire model:

- In a multiple linear regression such as this, significance is a test of whether the regression model is better than a model with no independent variables. In other words, is the model better than no model at all?
- Generally, a model is considered significant if the F statistic (Significance F in Illustration 8.29) is less than 0.05.

Purchasing Dept. Expenses ANOVA Results

ANOVA					
	df	SS	MS	F	Significance F
Regression	2	147659116.6	73829558.29	41.29197	5.2812E-08
Residual	21	37547755.8	1787988.371		
Total	23	185206872.4			

ILLUSTRATION 8.29 ANOVA Regression Statistics for Purchasing Department Expenses

So, is the model significant? The ANOVA in Illustration 8.29 has a Significance F of 5.2812E-08. The notation “E-08” after 5.2812 represents scientific notation, also known as exponential notation. 5.2812E-08 is the same as 0.000000052812. This is a number well below 0.05, so the model is significant. In other words, the independent variables can explain some variation of total expenses, so it is better than no model at all.

The last section of the regression summary output provides information to create the equation that predicts the dependent variable. If the adjusted R square and standard error are acceptable and the model is significant, then we can interpret the equation of the model.

The intercept and coefficients of the model represent the equation of the line that best fits the data. The key statistic to analyze in this section is the p -value for each of the independent variables. Like the F statistic, the p -value provides a test of significance. It is a test as to whether the independent variable improves the ability of the model to predict the dependent variable. A p -value of 0.05 or less is considered significant.

Let's use the output in **Illustration 8.30** to identify the prediction model for Super Scooters' purchasing department total costs, and then interpret the coefficients. Notice that the p -values for all the independent variables meet the test of being less than 0.05 and are therefore significant.

ILLUSTRATION 8.30 Regression
Model Example**Purchasing Dept. Regression Model**

	Coefficients	Standard Error	t Stat	P-value
Intercept	-994.572	877.1183137	-1.133908575	0.269612
# of Purchase Orders	180.0127	31.32217805	5.747132382	1.05E-05
Sales Volume	1.1914012	0.351035522	3.393961854	0.002737

The prediction model will be equal to the intercept, plus the coefficients of the independent variables, multiplied by the predicted values for those variables. Based on the regression model in Illustration 8.30, the equation to predict total purchasing department costs is:

$$(\$994.57) + \$180.01 (\text{number of purchase orders}) + \$1.19 (\text{sales volume})$$

Illustration 8.31 is the calculation of predicted total expenses if 12 purchase orders are processed and 2,200 scooters are sold.

ILLUSTRATION 8.31 Prediction
Model Example**Purchasing Dept. Predicted Expenses**

	Model Coefficient	Variable Values	Prediction
Intercept	\$ (994.57)	1	\$ (994.57)
# of Purchase Orders	\$ 180.01	12	\$ 2,160.15
Sales Volume	\$ 1.19	2,200	\$ 2,621.08
			\$ 3,786.66

Add the product of each independent variable coefficient and each variable's predicted value to the intercept for a total prediction of \$3,786.66 purchasing department costs for the year.

The model could be interpreted like this:

- **Intercept:** The intercept does not have a practical interpretation. It is a result of the model that represents the mean for the response when all the independent variables are zero. It is where the equation function crosses the y-axis.
- **# of Purchase Orders:** Each purchase order adds \$180.01 to total costs.
- **Sales Volume:** For each additional scooter sold, purchasing department costs increase by \$1.19.

Using a model like the one in Illustration 8.31 helps businesses predict future outcomes. The combination of evaluating the variables included in the model to see if they make sense and then assessing the regression model statistics will help determine if the model is both valid and reliable.

APPLYING CRITICAL THINKING 8.4: Interpret Predictive Analytics

If the analysis being interpreted predicts a future outcome, then you will be interpreting predictive analytics:

- In the Super Scooters' purchasing department costs example, the purpose of the model is to better understand and predict the department's costs (**Purpose**).
- Making this prediction requires understanding how to interpret a regression analysis (**Knowledge**).

Prescriptive Analytics

Prescriptive analytics prescribe what should happen to achieve desired results. The most common prescriptive analytics in accounting are what-if analyses and optimization models. The

same rules that apply to the other analytics methods also apply here. The analysis must be valid and reliable. Since a prescriptive model prescribes action, it is crucial to verify that inputs and outputs of the model are valid and reliable to avoid making poor business decisions.

A spreadsheet model that evaluates how changes to values and assumptions affect an outcome is called a **what-if analysis**. A what-if analysis is an easy way to change values in a spreadsheet and recalculate the outputs.

Excel tools that are commonly used for what-if analysis include Scenario Manager and Goal Seek. (Optimization models are discussed in the chapter about data analysis motivations and objectives.) Regardless of which tool is used, assessing the validity and reliability of the model output is the same:

- Understand what the model is addressing.
- Determine if the model measures what it is supposed to measure and if it does, in fact, represent the objective (validity).
- Review the model measures to confirm they are accurate and consistent (reliability).

What-If Analysis: Scenario Manager

Let's look at a scenario analysis and assess validity and reliability. An international dance company, Ballet Nuevo, is performing in Atlanta, Georgia. They have three performances scheduled and are determining if they should add a fourth. **Illustration 8.32** provides a summary model.

ILLUSTRATION 8.32 Ballet Performance Financial Model

	A	B	C
1	Ballet Nuevo-Additional Performance Scenario Analysis		
2			
3	Expected Ticket Sales	2,500	people
4	Average Concessions Spending	\$ 25.00	per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 50.00	per person
7	Concession Profit %	50%	
8	Profit Percentage	80%	
9			
10	Ticket Revenue	\$ 125,000.00	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 113,250.00	

Ballet Nuevo believes there are three possible scenarios, which are shown in **Illustration 8.33**.

Ballet Nuevo Scenario Analysis Model

	Likely	Optimistic	Pessimistic
Expected Ticket Sales	2,500	4,500	1,500
Concessions Spending	\$ 25.00	\$ 40.00	\$ 10.00
Theater Charges	\$ 18,000.00	\$ 10,000.00	\$ 25,000.00

ILLUSTRATION 8.33

Ballet Nuevo Performance Scenarios

A scenario analysis was performed using Scenario Manager in Microsoft Excel. The scenario summary shows the results when cells B3, B4, and B5 (Illustration 8.32) are changed to the values shown in Illustration 8.33.

The result of the analysis is shown in [Illustration 8.34](#).

ILLUSTRATION 8.34
 Ballet Nuevo Scenario Analysis

Scenario Summary				
	Current Values	Optimistic	Pessimistic	Likely
Changing Cells				
\$B\$3	2,500	4,500	1,500	2,500
\$B\$4	\$ 25.00	\$ 40.00	\$ 10.00	\$ 25.00
\$B\$5	\$ 18,000.00	\$ 10,000.00	\$ 25,000.00	\$ 18,000.00
Result Cells				
\$B\$13	\$ 113,250.00	\$ 260,000.00	\$ 42,500.00	\$ 113,250.00

The interpretation of this analysis is that even in the pessimistic scenario Ballet Nuevo will make a profit on an additional performance.

How do we know this analysis is valid and reliable? Ballet Nuevo would like to evaluate different ticket and concession sales scenarios to determine if they should add an additional performance. A scenario analysis is a valid method to use for this type of analysis, and the model represents three realistic possibilities. As to reliability of the model, that can be confirmed by verifying the profit calculation is correct and the assumptions are realistic (ticket prices, concession purchases, and theater costs). If the model inputs are accurate and consistent, then the model is reliable.

What-If Analysis: Goal Seek

Another tool for performing what-if analysis is Goal Seek:

- Goal Seek is used if the desired result is already known but the input value to achieve that result is not.
- Goal Seek is limited because it can only use one input variable. If the analysis being performed requires more than one variable to change, then an optimization model using Excel Solver would be needed.

In the previous example, Ballet Nuevo wanted an analysis comparing different ticket and concession sales scenarios to determine if they should add a performance. What if an additional performance is not possible due to theater availability, and Ballet Nuevo instead must consider how to set ticket prices to meet a specific target profit?

- Ballet Nuevo has determined they need a profit of \$150,000 from their performance.
- Currently the ticket prices are \$50, and they anticipate 2,500 ticket sales based on last year’s performances.

[Illustration 8.35](#) shows the financial information for Ballet Nuevo and the Goal Seek box that appears after clicking the **Data** tab in Excel and selecting **What-If Analysis** and **Goal Seek**.

ILLUSTRATION 8.35 Ticket Price and Target Profit – Goal Seek

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C
1	Ballet Nuevo – Profit Target Goal Seek		
2			
3	Expected Ticket Sales	2,500	People
4	Average Concessions Spending	\$ 25.00	Per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 50.00	Per person
7	Concession Profit %	50%	
8	Profit percentage	80%	
9			
10	Ticket Revenue	\$ 125,000.00	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 113,250.00	

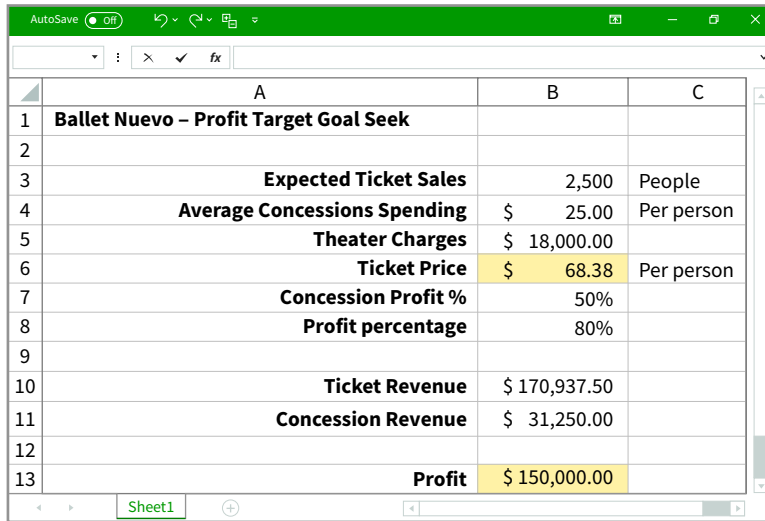
The Goal Seek dialog box is open, showing:

- Set cell:** \$B\$13
- To value:** 150000
- By changing cell:** \$B\$6

The dialog box has OK and Cancel buttons.

- The **Set cell** represents the profit calculation in cell B13 in the Excel spreadsheet.
- Note the current profit amount of \$113,250. The desired profit is \$150,000, so that value was entered in the **To value** box.
- The variable being manipulated is ticket price, so the cell reference for the ticket price (B6) was entered in the **By changing** cell box.

After the user clicks **OK**, Excel will calculate the ticket price necessary to meet the profit goal of \$150,000. **Illustration 8.36** is the solution generated by Excel. To meet a profit target of \$150,000, Ballet Nuevo must charge \$68.38 per ticket.



	A	B	C
1	Ballet Nuevo – Profit Target Goal Seek		
2			
3	Expected Ticket Sales	2,500	People
4	Average Concessions Spending	\$ 25.00	Per person
5	Theater Charges	\$ 18,000.00	
6	Ticket Price	\$ 68.38	Per person
7	Concession Profit %	50%	
8	Profit percentage	80%	
9			
10	Ticket Revenue	\$ 170,937.50	
11	Concession Revenue	\$ 31,250.00	
12			
13	Profit	\$ 150,000.00	

ILLUSTRATION 8.36 Ticket Price and Target Profit – Goal Seek Solution

To assess the reliability and validity of a goal seek model, determine if the model measures what it is supposed to measure (reliability) and if it represents the reality of the question/objective (validity):

- In this model, confirm the profit calculation is accurate.
- Additionally, determine if the model answers the question of how much Ballet Nuevo should charge per ticket to achieve a profit of \$150,000 and if the model's suggested ticket price is realistic. If \$68.38 is unrealistic for a ticket to a performance, then Ballet Nuevo must consider other ways to reach profit goals.

APPLYING CRITICAL THINKING 8.5: Interpret Prescriptive Analytics

Use the elements of critical thinking when interpreting a prescriptive model:

- Knowing who will use predictions helps determine if the model addresses their concerns and confirm that it represents reality (**Stakeholders**).
- An understanding of why the analysis is being performed is required to evaluate if the variables make sense (**Purpose**).
- Knowing how to interpret regression analyses and evaluate optimization models is necessary to interpret them (**Knowledge**).

Apply It 8.5

Interpret Regression Results



Data Managerial Accounting DHI would like to understand what is driving total expenses for the hotel chain. Luciana feels strongly that the following variables have the most influence on expenses:

- Age of the hotel
- The number of maintenance employees
- Total housekeeping hours
- Total rooms rented

She has asked you to prepare a regression model using those variables to predict expenses. The following illustration is the result of that model.

Housekeeping Hours and Rooms Rented by Location Regression Model

Regression Statistics					
Multiple R	0.7215151				
R Square	0.520584				
Adjusted R Square	0.4714132				
Standard Error	93901.466				
Observations	44				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	3.73412E+11	9.3353E+10	10.58724498	6.62556E-06
Residual	39	3.43882E+11	8817485253		
Total	43	7.17293E+11			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	413314.47	107148.5543	3.85739665	0.000418405	
Room Rentals	11.015461	2.918679199	3.77412543	0.000534577	
Hours Worked, Housekeeping	6.0934507	5.531393108	1.1016123	0.277382666	
Age	-3565.695	1674.938087	-2.12885155	0.039637083	
Employees, Maintenance	16762.392	6906.902129	2.42690446	0.019949355	

1. What does the adjusted *R* square reveal about the model?
2. Is the model better than not having a model?
3. Are there any variables you would recommend Luciana remove from the model? Why?

SOLUTION

1. The adjusted *R* square is 0.471, meaning that room rentals, housekeeping hours worked, the age of the hotel, and the number of maintenance employees can explain 47.1 % of total expenses.
2. The model is significant – Significance *F* is less than 0.05 – so, it is better than no model at all.
3. The *p*-value for Hours Worked, Housekeeping is greater than 0.05, so it should be removed from the model.

Chapter Review and Practice

Learning Objectives Review

1 Compare data analysis interpretation and data exploration.

While data exploration and interpretation seem similar, there are important differences:

- Data exploration is the process of analyzing data to determine whether we need to perform additional analyses. The objective in data exploration is getting to a point where we are sufficiently confident we understand what is happening in the data.

- Data analysis interpretation is the process of evaluating an analysis to understand it and explain its meaning. The insights gained from interpretation lead to good business decisions.

2 Apply critical thinking to data analysis interpretation.

- Understanding the stakeholders enables understanding the context of the analysis and the implications of the results.
- Identifying the purpose of the analysis maintains focus on its goal and avoids misdirection in the interpretation.

- When interpreting an analysis, consider if there are alternative explanations or alternative analyses that should be conducted.
- Identify potential risks such as data risks, analysis risks, and risks of bias.
- All analysis interpretation requires specific knowledge. Identifying the necessary accounting, industry, and technology knowledge gives us the tools to interpret the analysis.
- Similar analysis interpretation performed in the past could be applied in the current context.

3 Determine if the data analysis results answer the question and align with the objective of the analysis.

The answers to some specific questions can be helpful when interpreting a data analysis:

- Is the method and the result reasonable given current knowledge about the subject being analyzed? Does the analysis and its interpretation make sense? Consider whether the analysis has clear meaning. Does it address the objective or question posed? Consider also whether the interpretation is reasonable.
- Sometimes more information or analysis is needed before a final interpretation can be completed. Avoid risks like the “what you see is all there is” bias or confirmation bias.

4 Evaluate the validity and reliability of descriptive and diagnostic data analysis results.

Descriptive analytics determine what has happened in the past, while diagnostic analytics investigate why it happened.

- Common descriptive analytics techniques include frequency distributions, cross tabulation, measures of location, and measures of dispersion.

- Common diagnostic analytics techniques include anomaly detection, correlation analysis, and trend analysis.
- A data analysis is valid if it measures what it is supposed to measure and represents reality.
- A data analysis is reliable if the measures used in the analysis are accurate and consistent and the data used are dependable and trust worthy.

5 Evaluate the validity and reliability of predictive and prescriptive data analysis results.

Predictive analytics are used when the objective of the analysis is to predict a future outcome. Prescriptive analytics aim to prescribe actions that would result in best outcomes in the future.

- Linear regression is the foundation for most predictive modeling techniques. The validity of a regression model is assessed by how well the model represents the phenomenon of interest. Reliability of the model is assessed by evaluating the regression statistics, model statistics, and the p -value of the coefficients.
- Optimization models and what-if analysis are commonly used for prescriptive analytics. In the accounting profession, what-if analysis can help evaluate several choices (Scenario Analysis) or to determine a specific input (Goal Seek).
- Regardless of the tool used to create the what-if analysis, assess validity by determining if the model is measuring what it is supposed to measure (validity) and that the measures are accurate and consistent (reliability).

Key Terms Review

Accurate 8-17
Adjusted R square (R^2) 8-28
Anomaly 8-22
Correlation analysis 8-23
Correlation coefficient 8-23
Cross tabulation analysis 8-19
Data analysis interpretation 8-2
Frequency distribution 8-18

Linear regression 8-27
Mean 8-19
Median 8-19
Mode 8-19
Outlier 8-20
Regression statistics 8-28
Reliability 8-17
Scatterplot 8-20

Standard deviation 8-21
Standard error 8-28
Trend analysis 8-24
Trendline 8-24
Validity 8-17
what-if analysis 8-31

How To Walk-Throughs

HOW TO 8.1 Create a Frequency Distribution With Power BI

Illustration 8.14 is a frequency distribution for Super Scooters' different models. It was created in Excel, but other tools, such as Power BI, can also create frequency distributions.

What You Need: **Data** The How To 8.1 data file.

STEP 1: Extract the data. Open Power BI and select the **Home** tab on the top horizontal menu (**Illustration 8.37**).

How To



ILLUSTRATION 8.37 Extract and Load Super Scooters Data

HINT Be sure you check the box before "Sales Transactions 2024-2025." If you forget you will not have the option to "Load."

Sales Transactions 2024-2025

Order Number	Year	Sales Order Number	Model	Sold Date
13684	2024	13684-2024	Celeritas	12/31/2024
13685	2024	13685-2024	Lazer	12/31/2024
13682	2024	13682-2024	Captain	12/30/2024
13683	2024	13683-2024	Lazer	12/30/2024
13677	2024	13677-2024	Lazer	12/29/2024
13678	2024	13678-2024	Lazer	12/29/2024
13679	2024	13679-2024	Celeritas	12/29/2024
13680	2024	13680-2024	Captain	12/29/2024
13681	2024	13681-2024	Celeritas	12/29/2024
13675	2024	13675-2024	Celeritas	12/28/2024
13676	2024	13676-2024	Captain	12/28/2024
13671	2024	13671-2024	Captain	12/27/2024
13672	2024	13672-2024	Lazer	12/27/2024
13673	2024	13673-2024	Lazer	12/27/2024
13674	2024	13674-2024	Captain	12/27/2024
13668	2024	13668-2024	Captain	12/26/2024
13669	2024	13669-2024	Celeritas	12/26/2024
13670	2024	13670-2024	Kicks	12/26/2024
13665	2024	13665-2024	Lazer	12/25/2024

The data in the preview has been truncated due to size limits

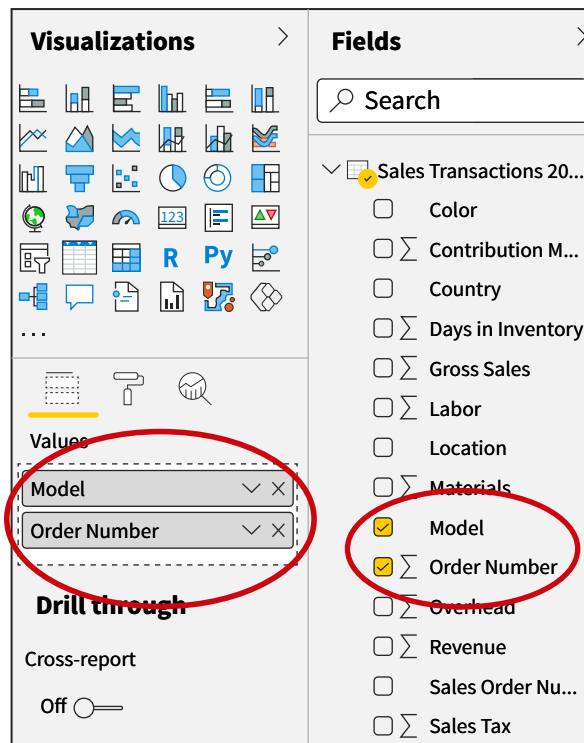
Load Transform Data Cancel

- Select the Excel icon below.
- When the file dialog box opens, navigate to the Super Scooters Excel file and select **Open** in the bottom right corner.

STEP 2: Load the data (Illustration 8.37). From Power BI's **Navigator** window, which automatically opens once the data source is selected, select the data for this exercise: Sales Transactions 2023–2025. Next, click **Load** on the bottom right to upload this data.

STEP 3: Transform the data to create a frequency table (Illustration 8.38). The screen will return to the main Power BI screen. The **Fields** column will be on the far right of the screen.

- Click the drop-down arrow next to Sales Transactions 2023–2025 to view all column names from the spreadsheet.
- Drag the data fields **Model** and **Σ Order Number** to the spaces provided directly under **Values**.
- Each field has a down arrow indicating a menu of options. Select this arrow for the **Σ Order Number** field and the pull-down menu appears.
- Select **Count**. The field name will change to **Count of Order Number**.
- On the left is a table with the results containing the four Super Scooter models, the count of order number, and a grand total. You may enlarge that new table to see all the columns.

ILLUSTRATION 8.38 Build a Frequency Table

STEP 4: Add a relative frequency table column by returning to the **Fields** column and selecting the **Order Number** field. Drag it below the **Count of Order Number** field in the **Visualizations** column.

- Select the arrow pull down menu for the new **Order Number** field and select **Count**.
- In the arrow pull down menu, choose **Show Value As**. Select the arrow to the right for **Percent of Grand Total**.
- The left of the screen will show a new column with the percentage of total sales for each scooter model and the grand total (**Illustration 8.39**). Again, it may be necessary to enlarge the table to see the new column.

Model	Frequency	Relative Frequency
Captain	1,010	27.71%
Celeritas	892	24.47%
Kicks	456	12.51%
Lazer	1,287	35.31%
Total Sales	3,645	100.00%

ILLUSTRATION 8.39 Final Super Scooters Frequency Table

- Finally, it is possible to change the names of the columns to “Frequency” and “Relative Frequency” by clicking the same down arrow used to select **Count** and instead choosing **Rename for this Visual**.

It is also possible to do a frequency distribution using PivotTables or by using data visualization software.

HOW TO 8.2 Calculate Descriptive Statistics in Microsoft Excel

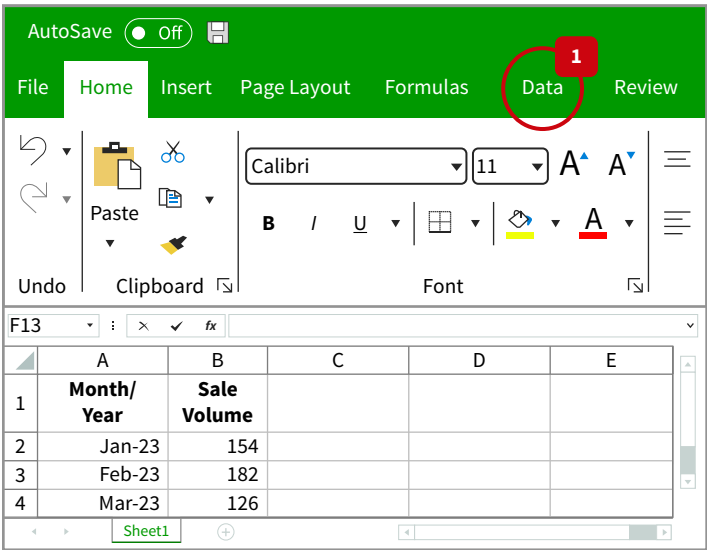
To create the Illustration of Super Scooters’ sales volume in Illustration 8.18, use the Descriptive Statistics option in the Data Analysis Tool in Microsoft Excel.



What You Need: **Data** The How To 8.2 data file.

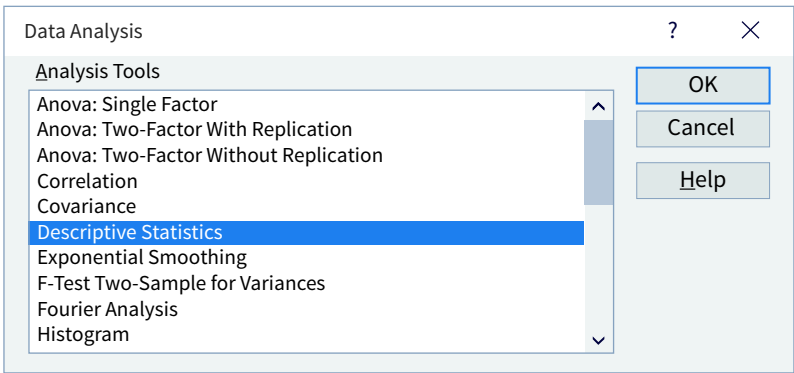
STEP 1: Open the worksheet by clicking **Data** in the toolbar. (Illustration 8.40).

ILLUSTRATION 8.40 Super Scooters Data



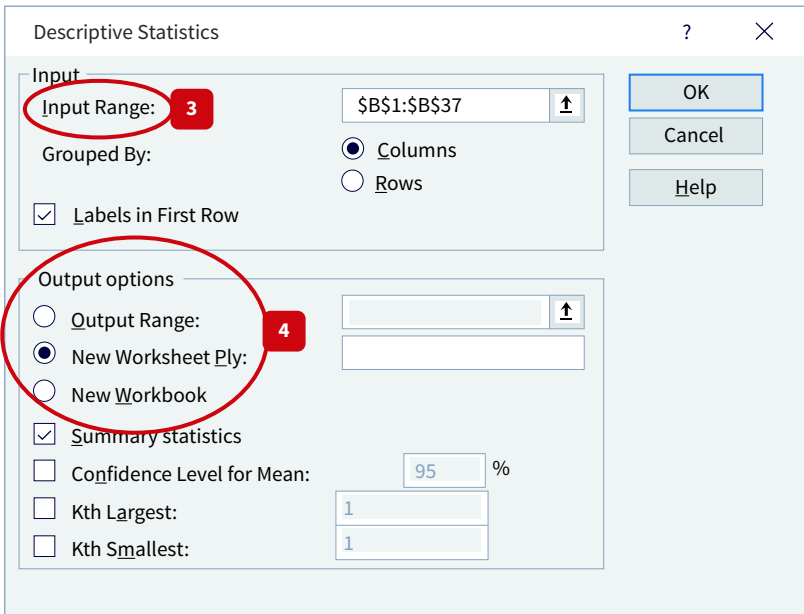
STEP 2: Select **Descriptive Statistics** (Illustration 8.41).

ILLUSTRATION 8.41 Analysis Tools Dialog Box



STEP 3: In the **Descriptive Statistics** box, designate the input range (Illustration 8.42).

ILLUSTRATION 8.42 Descriptive Statistics Box



- The input range is the data used to calculate descriptive statistics. In this example it is column B.
- Start with the column title and scroll to the bottom of the data set. (Hint: clicking in the title cell and holding Shift, Ctrl, and the keyboard down arrow will capture all the data in that column.)

STEP 4: Create the statistics output by clicking **Labels** and **Summary Statistics**.

Select **Output Range** and choose a cell in the current worksheet for the results, or click **New Worksheet** and Excel will put the results in a new workbook (Illustration 8.42).

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

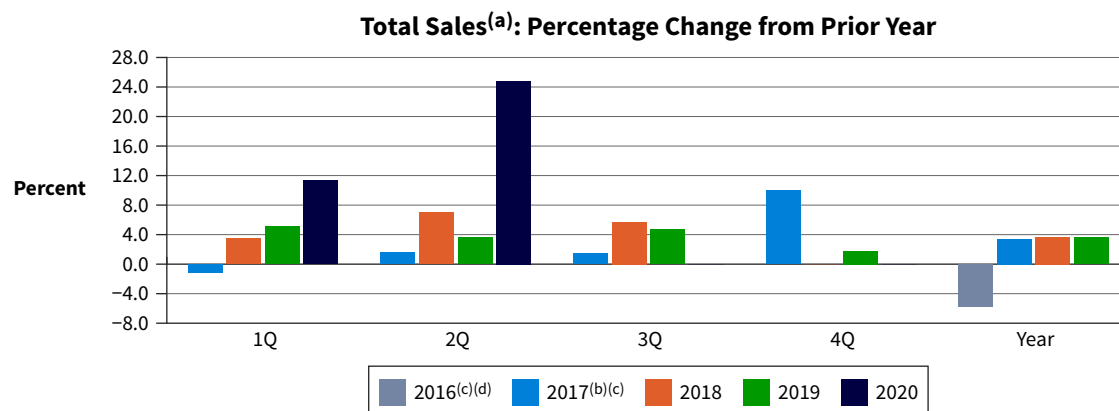
- (LO 1)** Data analysis results interpretation is the process of
 - evaluating an analysis to understand and explain its meaning.
 - performing descriptive analysis procedures to better understand the data.
 - exploring data to better understand relationships.
 - transforming data to enable analysis.
- (LO 1)** Which of the following is not a question addressed in data analysis results interpretation?
 - Were the correct data used to perform the analysis?
 - Was the correct technology used to perform the analysis?
 - Is the analysis biased?
 - Were the appropriate analysis methods used?
- (LO 1)** If the person preparing the analysis left out observations they did not think mattered to the decision, this would be an example of
 - data cleaning.
 - incorrect analysis methods.
 - fraud decision.
 - selection bias.
- (LO 1)** One of the most valued aspects of accountants is the ability to
 - be independent and skeptical evaluators of financial information.
 - perform difficult calculations.
 - identify employees committing fraud.
 - remember financial information.
- (LO 2)** To fully interpret data analysis results,
 - identify the person who prepared the analysis.
 - identify the purpose of the analysis.
 - prepare the analysis yourself.
 - have all the knowledge needed to understand the analysis.
- (LO 2)** When interpreting data analysis results,
 - consider only internal stakeholders affected by the results.
 - stakeholders are not relevant to the interpretation.
 - consider only external stakeholder affected by the results.
 - consider internal and external stakeholders potentially affected by the results.
- (LO 2)** Data analysis results that do not include the most recent data is an example of what type of risk?
 - Incorrect method of analysis
 - Preparer bias
 - Timeliness
 - Completeness
- (LO 2)** When interpreting data analysis results, which of the following could be a potential analysis risk?
 - Incorrect method
 - Missing relevant data
 - Data biases
 - Internal controls
- (LO 2)** The process of thinking about how past experiences can be applied to a current data analysis interpretation is which aspect of critical thinking?
 - Stakeholder identification
 - Alternatives
 - Self-reflection
 - Purpose
- (LO 3)** An analysis prepared to support a predetermined belief is an example of
 - selection bias.
 - fraud.
 - confirmation bias.
 - affect bias.
- (LO 3)** Which of the following describes the process of determining whether results of an analysis are reasonable given the intended question or purpose of analysis?
 - Making sense
 - Checking totals
 - Statistical analysis
 - Preparing data visualizations
- (LO 3)** If the answer to the question "Does the analysis address the needs/concerns of the stakeholders?" is no, then
 - you must re-do the same analysis to see if you get different results.
 - it will likely be acceptable if the numbers are correct.
 - it is likely the stakeholders did not understand the problem.
 - it is likely that a different analysis is needed before you can interpret the results.
- (LO 3)** When comparing two variables with different measurement scales
 - a clustered column chart is best to visually compare the variables.
 - a clustered bar chart is best to visually compare the variables.
 - a dual axis chart is best to visually compare the variables.
 - a line graph is best to visually compare the variables.

14. (LO 4) When assessing the reliability of the data analysis results, evaluate whether the
- measures used are accurate and consistent.
 - measures used measure what they are supposed to.
 - data used are timely.
 - data used are reasonable.
15. (LO 4) An appropriate analysis to determine how many times an event has occurred would be
- a measure of location.
 - a measure of dispersion.
 - a frequency distribution.
 - linear optimization.
16. (LO 4) An anomaly is
- always an outlier.
 - an observation that deviates from what is normal or expected.
 - always eliminated.
 - an indication of fraud.
17. (LO 4) When examining the relationship between two variables, if one variable increases as the other variable decreases the relationship is
- a positive correlation.
 - a negative correlation.
 - uncorrelated.
 - perfectly correlated.
18. (LO 4) If the objective is to use historical data to identify patterns, which is the best analysis to use?
- Linear optimization
 - Frequency distribution
 - Trend analysis
 - Linear regression
19. (LO 5) Which of the following analyses can predict a future outcome?
- Standard deviation
 - Linear optimization
 - Crosstabulation analysis
 - Linear regression
20. (LO 5) The goal of predictive analytics is to build a model that
- can help predict or better understand a phenomenon in which you are interested.
 - can be performed using statistical software.
 - can identify how frequently a phenomenon has happened in the past.
 - is not too complicated.
21. (LO 5) Assessing the reliability of regression analysis involves
- comparing the mean and median of the dependent variable.
 - making sure the model answers the question or purpose of the analysis.
 - reviewing the model statistics.
 - using what-if scenarios.
22. (LO 5) In a regression model prepared to predict revenue, which of the following is the correct interpretation of an adjusted R -square of 0.85?
- Revenue will increase by 85% next year.
 - The independent variables in the model can explain 85% of the change in revenue.
 - The dependent variable in the model can explain 85% of the change in independent variables.
 - The adjusted R -square is too small of a number for us to rely on the model.
23. (LO 5) A spreadsheet model that allows evaluating how changes to values and assumptions affect an outcome is called a
- regression equation.
 - linear optimization model.
 - best guess model.
 - what-if analysis.
24. (LO 5) Which is the best tool when the desired result is known, but not the input value for a single variable will achieve that result?
- Goal Seek
 - Scenario Manager
 - Linear regression
 - Data analysis

Review Questions

- (LO 1) Compare and contrast data exploration with data analysis interpretation.
- (LO 1) Discuss the two overall steps in the process of interpreting a data analysis.
- (LO 2) When applying the SPARKS critical thinking model to interpreting data analyses, how does identifying the stakeholders help interpret that analysis?

4. (LO 2) The following visualization was published on the Target Investors Relations website.



Fiscal Year	1Q	2Q	3Q	4Q	Year
2020	11.3 %	24.8 %			
2019	5.1 %	3.6 %	4.7 %	1.8 %	3.6 %
2018	3.5 %	7.0 %	5.7 %	- %	3.7 %
2017 (b) (c)	(1.1)%	1.6 %	1.4 %	10.0 %	3.4 %
2016 (c) (d)					(5.8)%

^(a) Total sales include merchandise sales, net of expected returns, from our store and digital channels, as well as gift card breakage.

^(b) The fourth quarter and full year 2017 consisted of 14 weeks and 53 weeks, respectively, compared with 13 weeks and 52 weeks in the comparable periods presented.

^(c) Beginning with the first quarter 2018, we adopted the new accounting standards for revenue recognition, leases, and pensions. We are presenting certain prior period results on a basis consistent with the new standards and conformed to the current period presentation. We provided additional information about the impact of the new accounting standards on previously reported financial information in a Form 8-K filed on May 11, 2018.

^(d) 2015 sales includes \$3,815 million related to our former pharmacy and clinic businesses, which Target sold to CVS in December of 2015.

Source: Target's Consolidated Financial Statements as filed with the U.S. Securities and Exchange Commission Public Domain.

Assume the role of the preparer of this visualization.

1. Who are the stakeholders?
2. What is the purpose of the analysis?
3. What applicable knowledge do you need in order to prepare this visualization?

Then, assume the role of the reviewer of this visualization and answer the following:

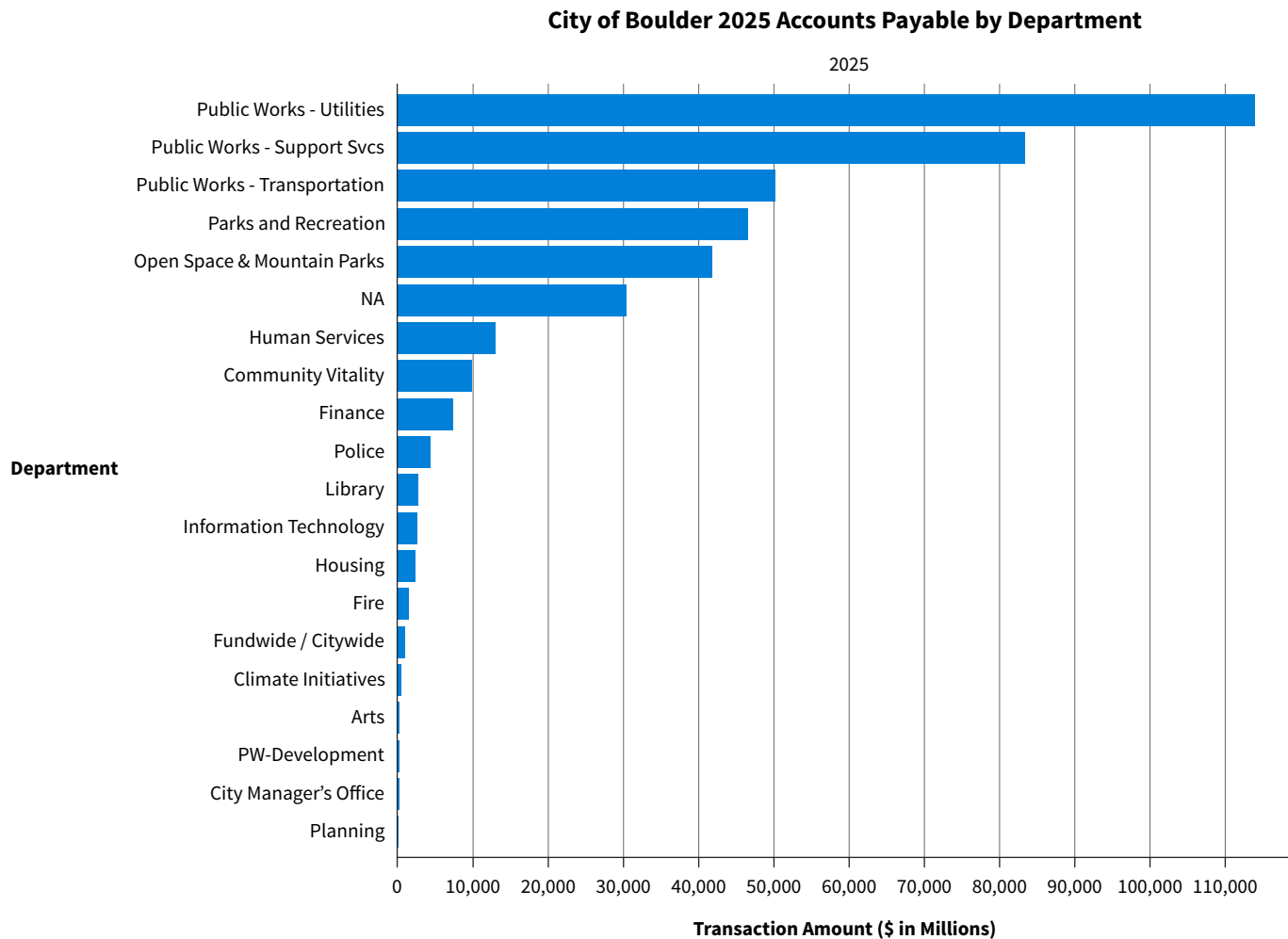
4. What are the risks when interpreting this visualization?
5. What conclusions can be identified from the visualization?
5. (LO 2) Explain why it is important to identify and evaluate alternative explanations of results.
6. (LO 3) Discuss what it means to ask "does the analysis make sense?"
7. (LO 3) What questions should be asked when evaluating the data and methods of an analysis?
8. (LO 3) What questions should be asked when evaluating the results of an analysis?
9. (LO 4) Explain what reliability and validity mean in the context of data analysis interpretation.
10. (LO 4) Explain the difference between an anomaly and an outlier.
11. (LO 5) Discuss how to evaluate a regression model for validity.
12. (LO 5) Explain how adjusted *R*-square can be used to evaluate the reliability of a regression model.
13. (LO 5) Explain how to interpret whether an independent variable in a regression model is a reliable measure of the dependent variable.

Brief Exercises

BE 8.1 (LO 1) While data exploration and data interpretation are related in the data analysis process, there is a distinction between these two activities. Data exploration is about understanding _____, whereas data interpretation is about understanding the _____.

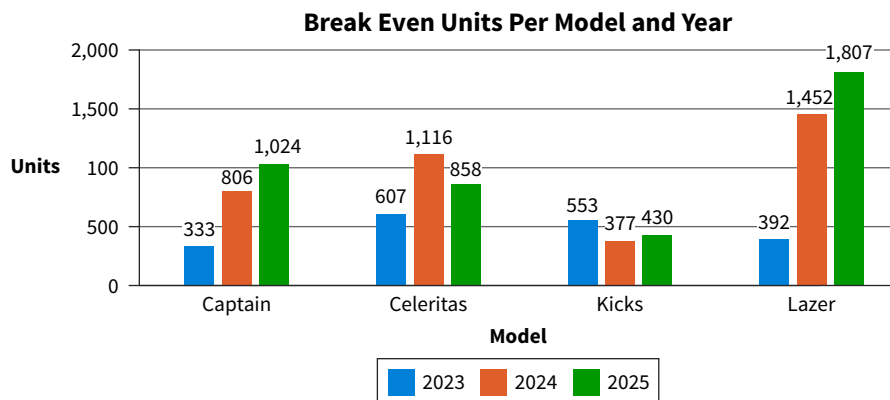
BE 8.2 (LO 1) There are steps in the process of reviewing an analysis. The first asks if the analysis makes sense, and the second asks if the analysis results are _____ and _____.

BE 8.3 (LO 1) Financial Accounting You are a financial analyst working for the city of Boulder, Colorado. Your manager has asked you to review the account payable analysis prepared by the accounts payable department. The analysis preparer told you the analysis is intended to determine which Boulder, Colorado city department has the highest level of spending in U.S. dollars during the year 2025. The analysis was prepared using all the transaction data for the year 2025 for every department. The preparer also verified that the total agreed to the appropriate line item for expenditures in the general ledger, and that there were no intercompany transactions to eliminate.



1. The original question for the analysis was “Which department has the highest spending amount during year 2025?” Does the analysis answer this question? Why or why not?
2. Based on the information from the analysis preparer, was the correct data used in the analysis? Why or why not?
3. What information in the analysis preparer’s statement allows you to conclude that the results of the analysis are accurate?

BE 8.4 (LO 2) Financial Accounting Managerial Accounting As a senior financial analyst at Super Scooters, you received the following graph from a financial analyst in your department. Your manager would like to know the quantity of units that must be sold for each model before it will have a profit.



The preparer of the graph told you that total sales price, variable costs, and total fixed costs were pulled from the accounting information system and that the amounts agreed to the general ledger. They used the break-even point formula from the CPA exam to calculate the break-even point in units for each model.

- What accounting and other knowledge would you need to obtain or apply in order to appropriately interpret the prepared graph or the statement made by the preparer?
- Who are the stakeholders for this analysis?
- What is the purpose of the analysis, and does the graph answer the question posed by your manager?

BE 8.5 (LO 2) When considering possible risks of a data analysis, always ask questions about the analysis. Match each type of risk to the question that addresses it. Each be used once, more than once, or not at all.

- | | |
|----------------------------|---------------------------------|
| a. Potential data risk | d. Potential visualization risk |
| b. Potential analysis risk | e. Potential user risk |
| c. Potential bias risk | |

Question	Type of Risk Addressed
1. Did the preparer have any potential biases that could have influenced the preparation of the analysis?	
2. Are the data used in the analysis the most recent available?	
3. Were all the necessary and appropriate data included in the analysis?	
4. Were appropriate internal controls in place to ensure the data used was correct?	
5. Was the correct method used to perform the analysis?	
6. Is the analysis missing relevant data?	
7. Did the analysis meet the original objective and answer the question?	

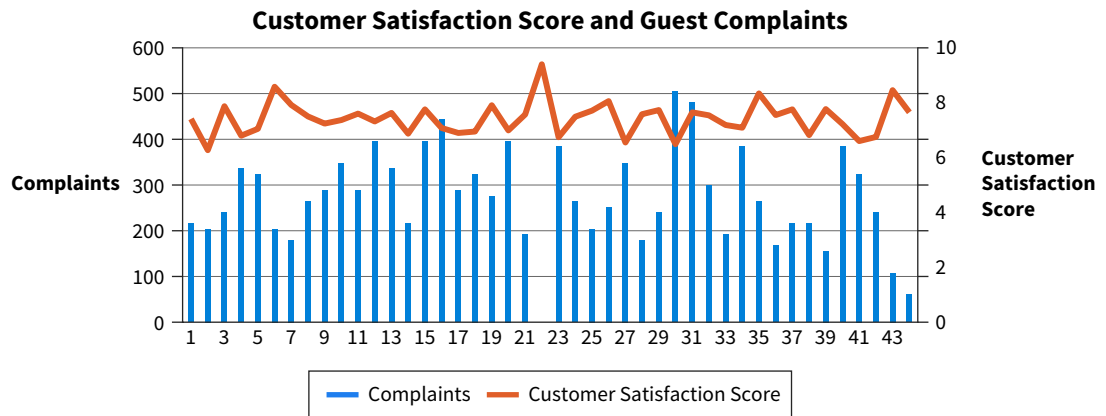
BE 8.6 (LO 3) Match each statement to the appropriate bias to be aware of when interpreting data analyses. Choose from the following biases. Each can be used once or more than once.

- | | |
|----------------------|-------------------|
| a. Confirmation bias | b. Selection bias |
|----------------------|-------------------|

Statement	Type of Bias
1. The person performing the analysis wants to prove a predetermined assumption.	
2. The person performing the analysis selected the data subjectively.	
3. The person interpreting the analysis wants to prove a predetermined assumption.	
4. The person interpreting the analysis focuses on results that support the existing assumption.	
5. The person interpreting the analysis considers only a sample of the data rather than the entire population.	
6. The person interpreting the analysis ignores aspects of the analysis that contradicts the existing assumption.	

BE 8.7 (LO 3) Auditing Roberto is the operations manager for Denton Hospitality Co. He has asked you to interpret the analysis of guest complaints. Internal audit provided the following analysis and information:

“We prepared a dual axis graph to show the relationship between guest complaints and customer satisfaction. The x-axis shows the hotel properties. The left y-axis is the number of complaints, and the right y-axis is the customer satisfaction score. The line across the graph is the customer satisfaction score for each hotel property. There is a strong correlation between customer satisfaction score and guest complaints.”



1. Roberto would like to know if guest complaints are related to maintenance expenses. Does the visualization and information provided by the internal audit department make sense given Roberto's question? Why or why not?
2. Roberto would like to know if there is a statistical correlation between customer satisfaction scores and guest complaints. Is the analysis sufficient to answer this question? Why or why not?
3. Roberto wants to present a visual depiction of the relationship between customer satisfaction score and guest complaints as part of a presentation to the board. Is the analysis method reasonable? Why or why not?

BE 8.8 (LO 4) Auditing You are an external auditor for U.S. Outdoor Adventure Stores. Your statistical analysis team has provided a table of descriptive statistics for the company's sales from 2022 to 2025.

U.S. Outdoor Adventure Stores	
Sales 2022–2025	
Mean	229.86
Median	54.49
Mode	12.96
Standard Deviation	623.25
Range	22,638.04
Minimum	0.44
Maximum	22,638.48
Count	9,994.00

The preparer of the analysis told you that they used all the sales transaction data and agreed total sales in the file to the sales line item in the general ledger. Note that there is a large difference between the mean and the median in the sales data set. What does a large difference between mean and median suggest about the sales transactions for U.S. Outdoor Adventure Stores from 2022 to 2025?

BE 8.9 (LO 4) Financial Accounting Service King, Inc. is a commercial cleaning and lawn care service company. Your manager wants to understand whether there is a relationship between the results of a customer satisfaction survey and key performance measures of each cleaning crew.

	Correlation: Customer Satisfaction Score
Customer Satisfaction Score	1
Annual Customer Complaints	–0.826124807
Minutes Spent at Property	0.938922087
Number of Employees	0.681057818

The preparer of the correlation analysis provided the data definitions:

- Customer Satisfaction Score: Score from the customer survey.
- Annual Customer Complaints: Number of customer complaints filed each year.
- Minutes Spent at Property: Number of minutes each cleaning crew spent cleaning a property.
- Number of Employees: Number of employees assigned to work at the property for a cleaning.

Use this information to match the appropriate answer to each question. Responses can be used once, more than once, or not at all.

- | | |
|--|--|
| a. A positive correlation coefficient. | e. There is a moderate positive correlation between the variables. |
| b. A negative correlation coefficient. | f. There is a moderate negative correlation between the variables. |
| c. There is a strong positive correlation between the variables. | g. There is no relationship between the variables. |
| d. There is a strong negative correlation between the variables. | |

Question	Answer
1. What does an inverse relationship mean between two variables?	
2. How would you interpret the correlation coefficient for annual customer complaints and customer satisfaction score?	
3. How would you interpret the correlation coefficient for minutes spent at the property and customer satisfaction score?	
4. How would you interpret the correlation coefficient for number of employees and customer satisfaction score?	

BE 8.10 (LO 5) Managerial Accounting Your client, All Care Hospital, needs to understand the cost drivers of their total hospital expenses. Your team ran regression analysis based on input from the hospital CFO. The regression equation follows:

Hospital expenses = (\$39,702) + \$564 (Beds) + \$0.65 (Outpatient Visits) + \$26.76 (Births)

Summary Output	
Regression Statistics	
Multiple R	0.87254918
R Square	0.76134208
Adjusted R Square	0.76098337
Standard Error	158,773.661
Observations	2,000

1. Identify the independent variables in the regression model.
2. Identify the dependent variable in the regression model.
3. Identify the regression statistic that indicates how well the model's independent variables explain the dependent variable.
4. What percentage of the variance in hospital expenses do beds, births, and outpatient visits explain?

BE 8.11 (LO 5) Tax Accounting One consideration of tax planning is understanding and projecting the amount of net revenue that a company expects for the upcoming year. Your team has created a what-if analysis to predict Super Scooters' net revenue for next year. Based on current year data, the team has performed three sensitivity analyses:

- Best-case scenario: They expect sales to increase by 10%.
- Likely-case scenario: They expect sales to increase by 5%.
- Worst-case scenario: They expect sales to decrease by 7%.

The current calculation of net revenue follows. Variable costs are 47% of sales, and Super Scooters expects variable costs to remain stable over the next year.

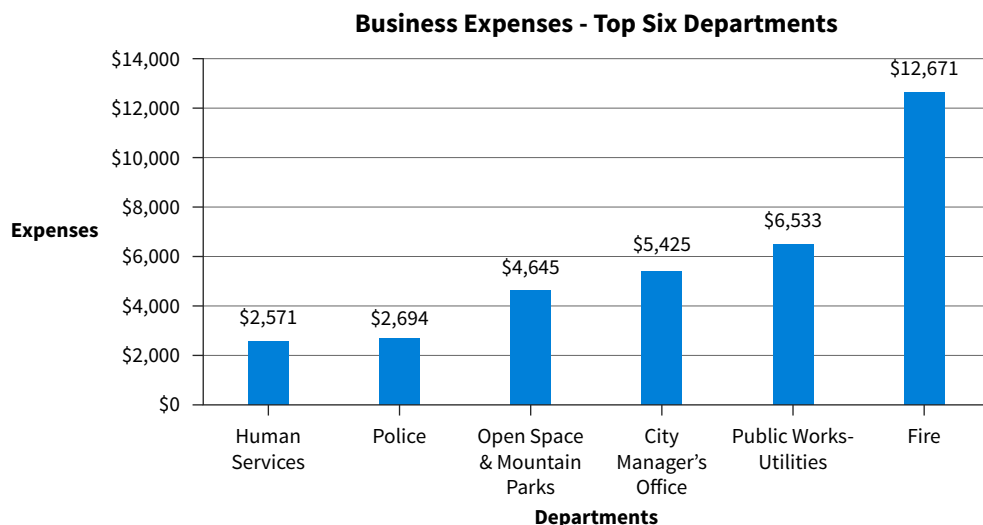
Net Revenue	
	2025
Gross Sales	\$ 12,165,162
Total Variable Costs	\$ 5,722,777
Contribution Margin	\$ 6,442,385
Total Fixed Costs	\$ 1,696,822
Net Revenue	\$ 4,745,563

What-If Analysis			
	Best-Case	Likely-Case	Worst-Case
Gross Sales	\$ 12,165,162	\$ 12,165,162	\$ 12,165,162
Total Variable Costs	\$ 6,008,916	\$ 5,837,232	\$ 6,123,371
Contribution Margin	\$ 6,156,246	\$ 6,327,930	\$ 6,041,791
Total Fixed Costs	\$ 1,696,822	\$ 1,696,822	\$ 1,696,822
Net Revenue	\$ 4,459,424	\$ 4,631,108	\$ 4,344,969

1. What knowledge would you need to understand how the best-case, likely-case, and worst-case scenarios should be interpreted?
2. Assume the objective of the analysis is to estimate sales tax for the upcoming year. Would the available what-if analysis provide the appropriate information to begin your estimation?

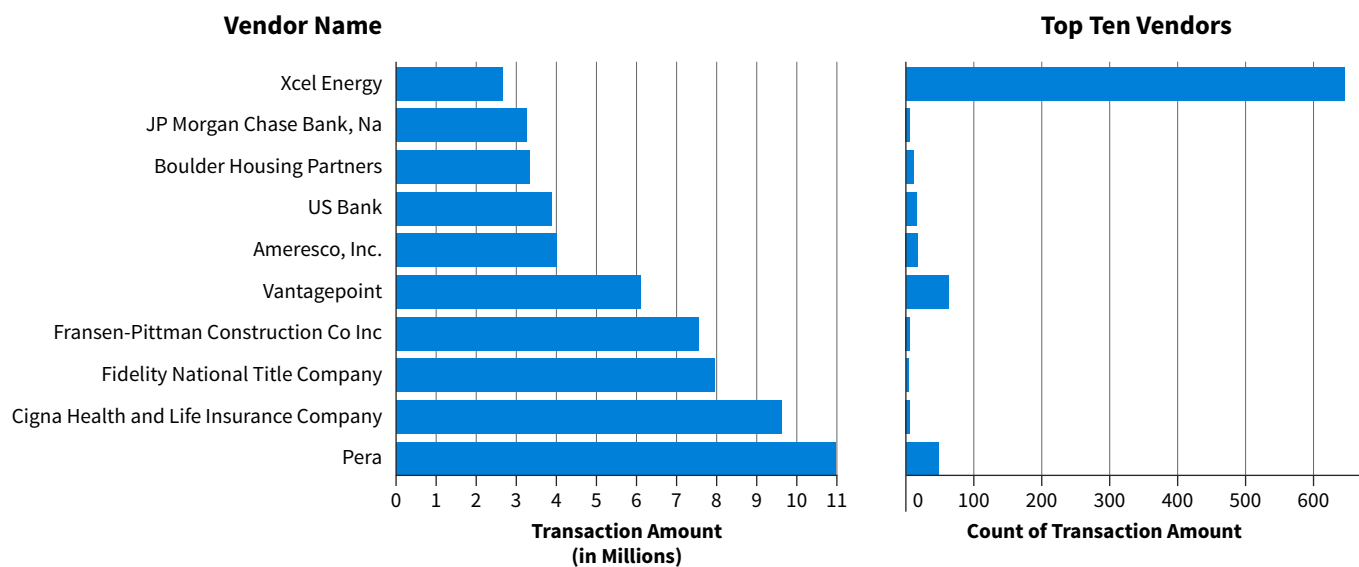
Exercises

EX 8.1 (LO 1) Auditing Accounting Information Systems Interpret and Evaluate a Data Visualization The following analysis was prepared to evaluate business expenses for the city of Boulder, CO. You work for the internal audit department for the city. Your manager asked you to review an analysis to evaluate business expense reimbursement for the city departments.



Answer the five data interpretation questions to determine if the analysis and visualization make sense and whether they are valid and reliable.

EX 8.2 (LO 1) Auditing Accounting Information Systems Interpret and Evaluate a Data Visualization An anonymous tip was called into the city's fraud hotline alleging that contracts are being awarded without the appropriate bidding process. An analysis was prepared to identify the top ten vendors so the bidding process for those vendors could be reviewed. The data used was all accounts payable for the current year. The vendors with the highest number of transactions were identified as top vendors.



Answer the five data interpretation questions to determine if the analysis makes sense and is valid and reliable.

EX 8.3 (LO 2) Accounting Information Systems Evaluate a Data Visualization Go to the Tableau Gallery website and search for an accounting topic. Choose a visualization and analyze it using the critical thinking framework.

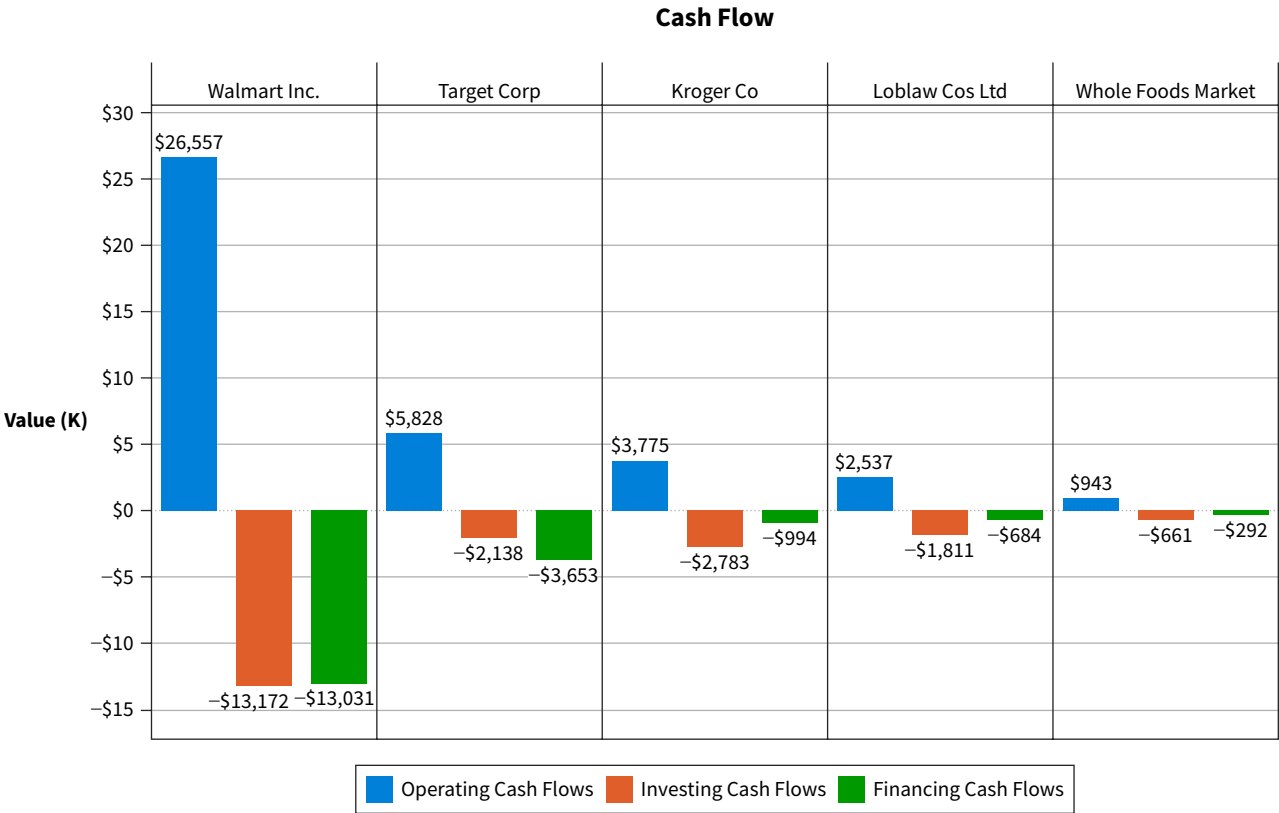
1. Who are the stakeholders ?
2. What was the purpose for analysis?
3. Are there alternative explanations or analyses?
4. Are there any risks that should be considered?
5. What knowledge did the preparer need and what knowledge does the interpreter need to understand the analysis?
6. How can you use your prior experience or apply this experience to future analyses?

EX 8.4 (LO 2) Financial Accounting Interpret Accounts Receivable Analyses Assume you are a financial analyst working in the controller's group at Super Scooters. You have been asked to analyze accounts receivable and have prepared the following aging table. The bars represent total accounts receivable balances.

Super Scooters, Inc. Accounts Receivable Aging Table					
Customer	Value				
C1036	\$40K OK	\$7,184	0	0	0
C1282	\$40K OK	0	0	0	\$32,478
C1423	\$40K OK	\$17,635	0	0	0
C1671	\$40K OK	\$23,038	0	0	0
C2036	\$40K OK	\$763	0	0	0
C2274	\$40K OK	\$23,562	0	0	0
C3168	\$40K OK	0	0	0	\$32,229
C9591	\$40K OK	\$10,840	0	0	0
C9917	\$40K OK	0	0	0	\$31,046
		Less than 30	31 to 60	61 to 90	91 to 180
		Days			

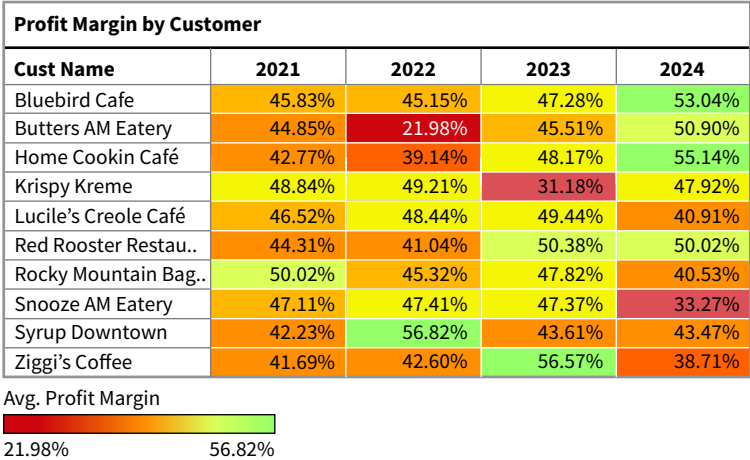
1. Who are the stakeholders?
2. What was the purpose for the analysis?
3. Are there alternative explanations or analyses?
4. Are there any risks that should be considered?
5. What knowledge did the preparer need, and what knowledge does the interpreter need to understand the analysis?
6. How can you use your prior experience or apply this experience to future analyses through self-reflection?

EX 8.5 (LO 2) Financial Accounting Interpret Cash Flow Visualizations You work as a financial analyst in the financial reporting group at Walmart, a large public retail company. You have been asked to interpret an analysis of the company’s cash flow compared to competitors, and you have been provided the following visualization.



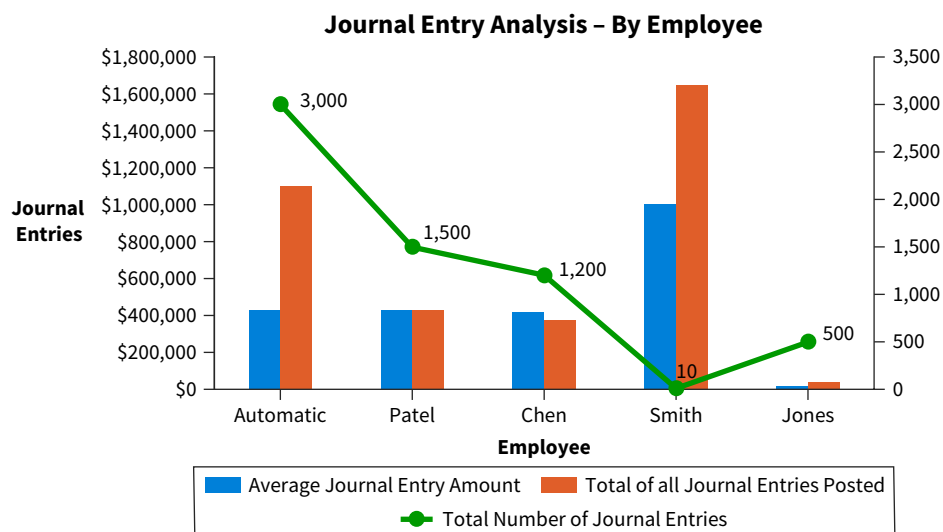
1. Who are the stakeholders ?
2. What was the purpose for analysis?
3. Are there alternative explanations or analyses?
4. Are there any risks that should be considered?
5. What knowledge did the preparer need, and what knowledge does the interpreter need to understand the analysis?
6. How can you use your prior experience or apply this experience to future analyses through self-reflection?

EX 8.6 (LO 3) Data Financial Accounting Interpret Reasonableness of a Data Analysis As a financial analyst at Best Bakes Bakery, you are often tasked with conducting and evaluating data for decision- making. The Best Bakes owners have asked you to identify their top five customers. You performed an analysis of profit margin by customer and prepared the following visualization using the Best Bakes data file.



1. Are the data used in the analysis reasonable given the question/objective of the analysis?
2. Is the analysis method reasonable given the question/objective of the analysis?
3. Are the results of the analysis reasonable given what you know about the subject being analyzed?
4. Are the implications of the analysis reasonable given what you know about the subject being analyzed?
5. Does the analysis address the needs/concerns of the stakeholders?

EX 8.7 (LO 3) Auditing Accounting Information Systems Interpreting Reasonableness of Journal Entries Data Analysis You are an audit associate responsible for the audit of journal entries for a public company client. Part of the audit procedures are analyzing journal entry activity to identify any unusual activity. You have prepared the following analysis of journal entry activity. It shows the total dollar amount, the average dollar amount, and the total number of journal entries made automatically by the accounting system and made by three key employees at your client.

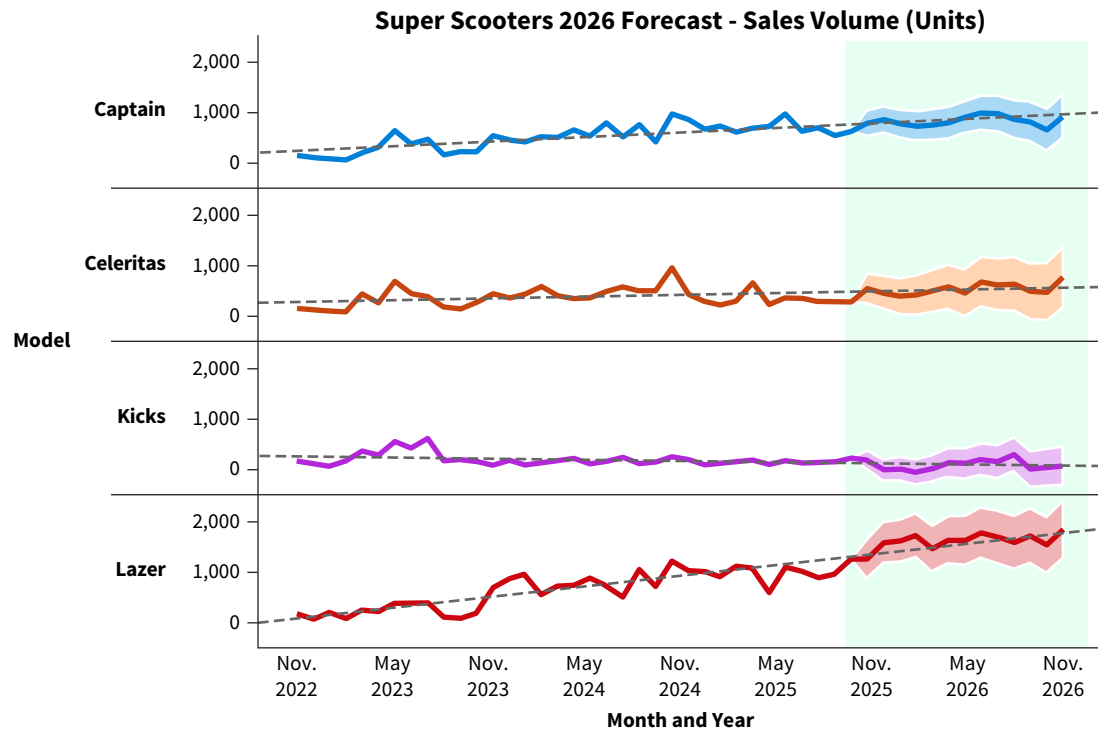


1. Are the data used in the analysis reasonable given the question/objective of the analysis?
2. Is the analysis method reasonable given the question/objective of the analysis?
3. Are the results of the analysis reasonable given what you know about the subject being analyzed?
4. Are the implications of the analysis reasonable given what you know about the subject being analyzed?
5. Does the analysis address the question/objective of the analysis?

EX 8.8 (LO 3) Managerial Accounting Interpreting Reasonableness of Sales Forecast
Analyses Super Scooters is considering dropping the Celeritas scooter model and expanding their electric-powered scooter offerings to take advantage of changing market demand. You acquired the following information:

- The market for electric scooters is growing rapidly due to the increase in scooter-sharing programs.
- Celeritas is the only gas-powered scooter that Super Scooters manufactures.
- The Lazer and Captain models are electric-powered scooters, whereas the Kicks model is a manually propelled scooter.

Your analytics team prepared this visualization to contribute to the decision-making process.



1. Are the data used in the analysis reasonable given the question/objective of the analysis?
2. Is the analysis method reasonable given the question/objective of the analysis?
3. Are the results of the analysis reasonable given what you know about the subject being analyzed?
4. Are the implications of the analysis reasonable given what you know about the subject being analyzed?
5. Does the analysis address the needs/concerns of the stakeholders?

EX 8.9 (LO 3) Data Financial Accounting Managerial Accounting Evaluate Data Analyses of Sales As a financial analyst for Super Scooters you must evaluate the following analysis prepared by the data analysis team using the Super Scooters data set. The analysis reports the sum of sales dollars for each model sold by the company. Based on the analysis, it appears that the Captain is the most popular model and the Kicks is the least popular model.

Model	Sum of Gross Sales
Captain	\$ 12,324,133
Celeritas	\$ 4,792,338
Kicks	\$ 1,108,183
Lazer	\$ 8,922,753
Grand Total	\$ 27,147,407

1. Does the analysis confirm the most and least popular models? Why or why not?
2. Do you think more analysis is needed?
3. If yes, what additional analyses would you like to see before deciding which model is the most popular?

EX 8.10 (LO 3) Data Managerial Accounting Evaluate Sales Trend Analysis Management at the U.S. Outdoor Adventure Company wants to understand which regions are performing best and if there is a trend in sales over the years within the regions. Your team has analyzed sales and prepared a visualization using the U.S. Outdoor Adventure Company Tableau file.

Yearly Region Sales				
Region	2022	2023	2024	2025
Central	\$103,838	\$102,874	\$147,429	\$147,098
East	\$128,680	\$156,332	\$180,529	\$213,239
South	\$103,846	\$ 71,301	\$ 93,539	\$122,977
West	\$147,883	\$139,766	\$186,976	\$250,633



 \$0 \$300,000

- Does the analysis address the objective? Why or why not?
- Do you think more analysis is needed?
- If yes, what additional analyses would you like to see before deciding which region is performing best?

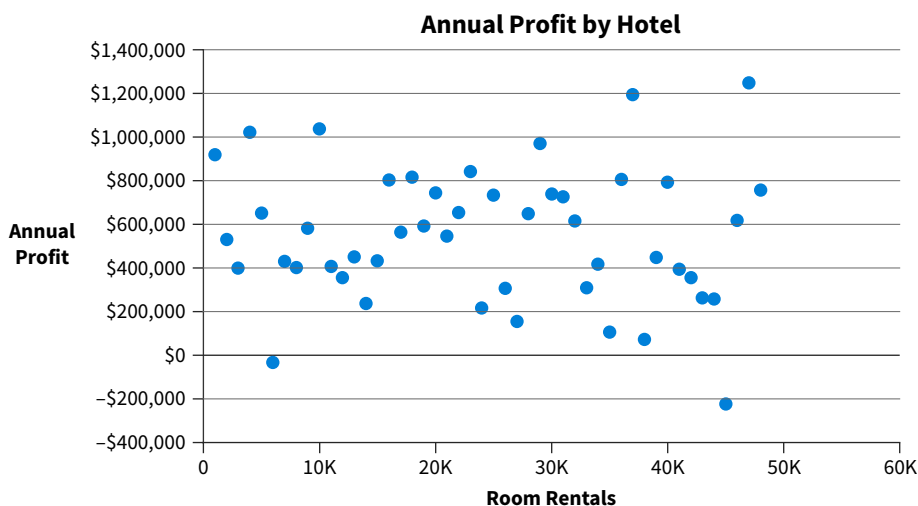
EX 8.11 (LO 4) Managerial Accounting Interpret Correlation Analysis All Care Hospital would like to better understand cost drivers of total expenses. As a first step they have prepared a correlation analysis to determine factors that are correlated with total expenses.

Cost Drivers	Beds	Outpatient Visits	Births	Gen Med / Surg or Not	Total Expense
Beds	1				
Outpatient Visits	0.61584604	1			
Births	0.66665244	0.496333644	1		
Gen Med / Surg or Not	0.25764602	0.322183655	0.359453	1	
Total Expense	0.77739236	0.782692108	0.62695	0.249571064	1

Interpret the correlations:

- Outpatient visits and total expense
- Number of beds and total expense
- Number of beds and births

EX 8.12 (LO 4) Financial Accounting Interpret Descriptive Statistics You are a financial analyst at the hotel chain Denton Hospitality. You have been asked to provide a better understanding of their profit from the past year. Your data analysis team has prepared two analyses. The first is a scatterplot that shows the total annual profit for each hotel. The second is a descriptive statistics analysis of profit from the past year.



Profit	
Mean	547,691.05
Standard Error	44,607.48
Median	553,868.00
Mode	#N/A
Standard Deviation	309,049.66
Sample Variance	95,511,689,341.06
Kurtosis	0.07
Skewness	0.00
Range	1,472,109.00
Minimum	(224,305.00)
Maximum	1,247,804.00
Sum	26,289,170.25
Count	48.00

- Interpret the following descriptive statistics:
 - Mean
 - Median
 - Standard Deviation
- Does the scatterplot support your explanation of the descriptive statistics? Why or why not?

EX 8.13 (LO 4) **Data** **Financial Accounting** **Managerial Accounting** **Interpret Descriptive Analytics** You are a financial analyst for One Stop Shop, which is a wholesale distributor of consumer products to convenience stores. The company operates in Canada, Mexico, and the United States. One Stop Shop has two sales channels:

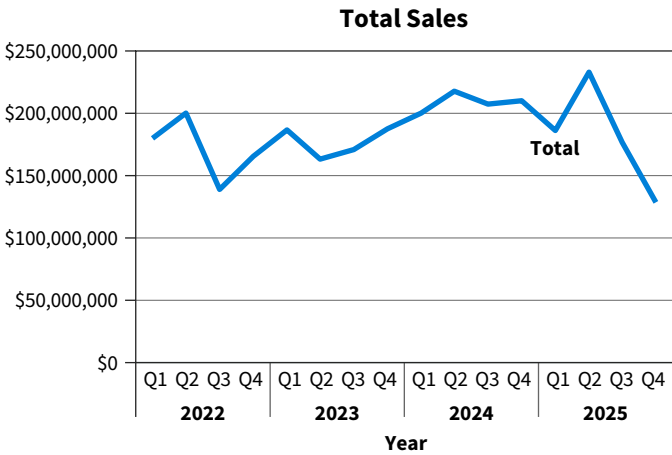
- Online sales made through the One Stop Shop website.
- Offline sales made directly with One Stop Shop sales representatives.

One Stop Shop would like to evaluate whether they should move to only online sales. They believe this will save a significant amount of money since they will not need as many sales representatives and will not have to pay sales commissions. Following is an analysis prepared by One Stop Shop using the One Stop Shop data set.

Row Labels	Sum of Total Sales
Canada	\$ 788,914,146
Offline	\$ 392,731,724
Online	\$ 396,182,421
Mexico	\$ 710,979,369
Offline	\$ 400,360,732
Online	\$ 310,618,637
United States	\$ 1,452,693,443
Offline	\$ 720,489,036
Online	\$ 732,204,407
Grand Total	\$ 2,952,586,958

1. What information does the pivot table provide to One Stop Shop?
2. What additional analyses should One Stop Shop consider?

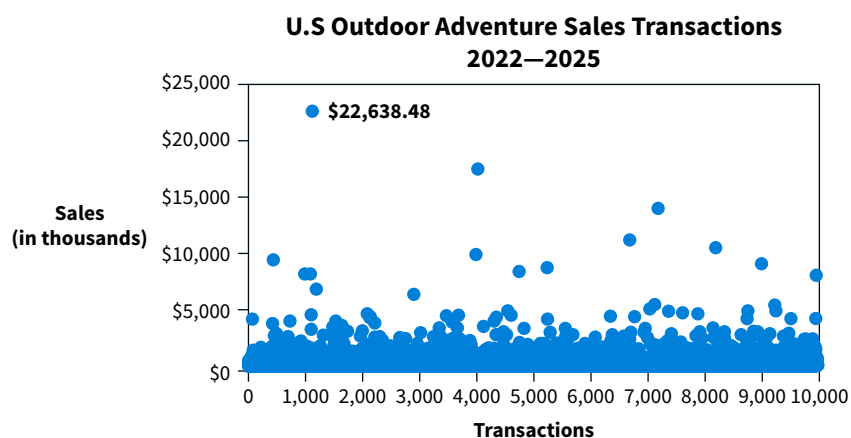
EX 8.14 (LO 4) **Financial Accounting** **Managerial Accounting** **Interpret Diagnostic Analytics** The management team at One Stop Shop wants to know if there is a seasonal pattern to sales. The data analysis team prepared the following visualization.



1. Does the analysis address the objective? Why or why not?
2. What additional analyses could be performed to identify sales patterns?

EX 8.15 (LO 4) Data Auditing Interpret Descriptive Analytics Your data analysis group has given you, an external auditor auditing sale transactions for U.S. Outdoor Adventure stores, an analysis that includes the standard deviation for sales from 2022 to 2025. They also gave you a scatterplot of sales.

Descriptive Statistics	
Sales 2022 — 2025	
Mean	229.86
Median	54.49
Mode	12.96
Standard Deviation	623.25
Sample Variance	388,434.46
Minimum	0.44
Maximum	22,638.48
Count	9,994.00



The preparer of the analysis used all the sales transaction data and agreed total sales in their file to the sales line item in the general ledger.

1. Interpret the variance and standard deviation for U.S. Outdoor Adventure Stores sales.
2. Does the scatterplot support your explanation of standard deviation? Why or why not?
3. Use the provided data file to recreate the analyses.

EX 8.16 (LO 5) Financial Accounting Managerial Accounting Interpret Regression Analysis Denton Hospitality Inc. has hired your consulting firm to evaluate and manage cashflows. Specifically, you have been asked to build a valid and reliable model to predict total expenses so management can better manage cashflows. You and your team prepared the following regression.

Hospitality, Inc – Total Expenses Prediction Model					
SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.810435721				
R Square	0.656806059				
Adjusted R Square	0.62160668				
Standard Error	79,448.54158				
Observations	44				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	4	4.71E+11	1.18E+11	18.65959	1.21E-08
Residual	39	2.46E+11	6.31E+09		
Total	43	7.17E+11			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	(131,136.43)	121071.9	−1.08313	0.285403	
Number of rooms	2,670.21	587.7595	4.543032	5.23E−05	
Hours Worked, Front Desk	29.30	9.270823	3.160299	0.003044	
Hours Worked, GM	52.01	18.93043	2.747581	0.009043	
Hours Worked, Housekeeping	13.41	4.683324	2.862522	0.006728	

1. Interpret the regression statistics.
2. Explain whether the model is significant.
3. What are the coefficients of the model?
4. Do the variables used in the model make sense? Why or why not?
5. Identify other variables that may improve the model.
6. Use the model to predict expenses for a hotel with the following characteristics:
 - Opened in 1975
 - 150 rooms
 - 9,200 hours worked – Front Desk
 - 1,500 hours worked – GM
 - 12,100 hours worked – Housekeeping
 - Location = Airport

EX 8.17 (LO 5) Data Managerial Accounting Interpret Sensitivity Analysis Super Scooters is preparing revenue forecasts for the next year. Based on current year data, you performed three sensitivity analyses:

- Best-case scenario: Variable costs will decrease by 5%.
- Likely-case scenario: Variable costs will increase by 2%.
- Worst-case scenario: Variable will increase by 7%.

The current calculation of net revenue is provided.

	2025
Gross Sales	\$ 12,165,162
Total Variable Costs	\$ 5,722,777
Contribution Margin	\$ 6,442,385
Total Fixed Costs	\$ 1,696,822
Net Revenue	\$ 4,745,563

The best-case, likely-case, and worst-case scenario analysis is provided. Super Scooters expects sales to remain stable over the next year.

	Best-Case	Likely-Case	Worst-Case
Gross Sales	\$ 12,165,162	\$ 12,165,162	\$ 12,165,162
Total Variable Costs	\$ 6,008,916	\$ 5,837,232	\$ 6,123,371
Contribution Margin	\$ 6,156,246	\$ 6,327,930	\$ 6,041,791
Total Fixed Costs	\$ 1,696,822	\$ 1,696,822	\$ 1,696,822
Net Revenue	\$ 4,459,424	\$ 4,631,108	\$ 4,344,969

Recreate the what-if analysis and answer the questions:

1. What is the question/objective of the analysis?
2. Does the model measure what it is supposed to measure?
3. Are the model measures accurate and consistent?

EX 8.18 (LO 5) Data Managerial Accounting Interpret Goal Seek Analysis U.S. Outdoor Adventure Sales has asked your accounting team to determine the break-even sales for their tent product. The team discussed potential methods to estimate break-even sales, and determined that Goal Seek is the most effective method. The sales team provided you with price and costs information:

- Sales price = \$70
- Variable cost per unit = \$25
- Fixed costs = \$10,000

With the spreadsheet model used to calculate profit using this data, use Goal Seek to answer the following questions.

1. What are the number of break-even units needed?
2. How many tents must U.S. Outdoor Adventures sell to make a profit of \$100,000?
3. How would you verify the validity and reliability of the model and model results?

EX 8.19 (LO 5) Data Financial Accounting Interpret Predictive Analytics You are a financial analyst at Best Bakes Bakery. Your manager has asked you to build a model to predict sales for 2026 using sales data from 2022 to 2025. The following is the trendline equation using the variables gross sales in USD and month of sale.

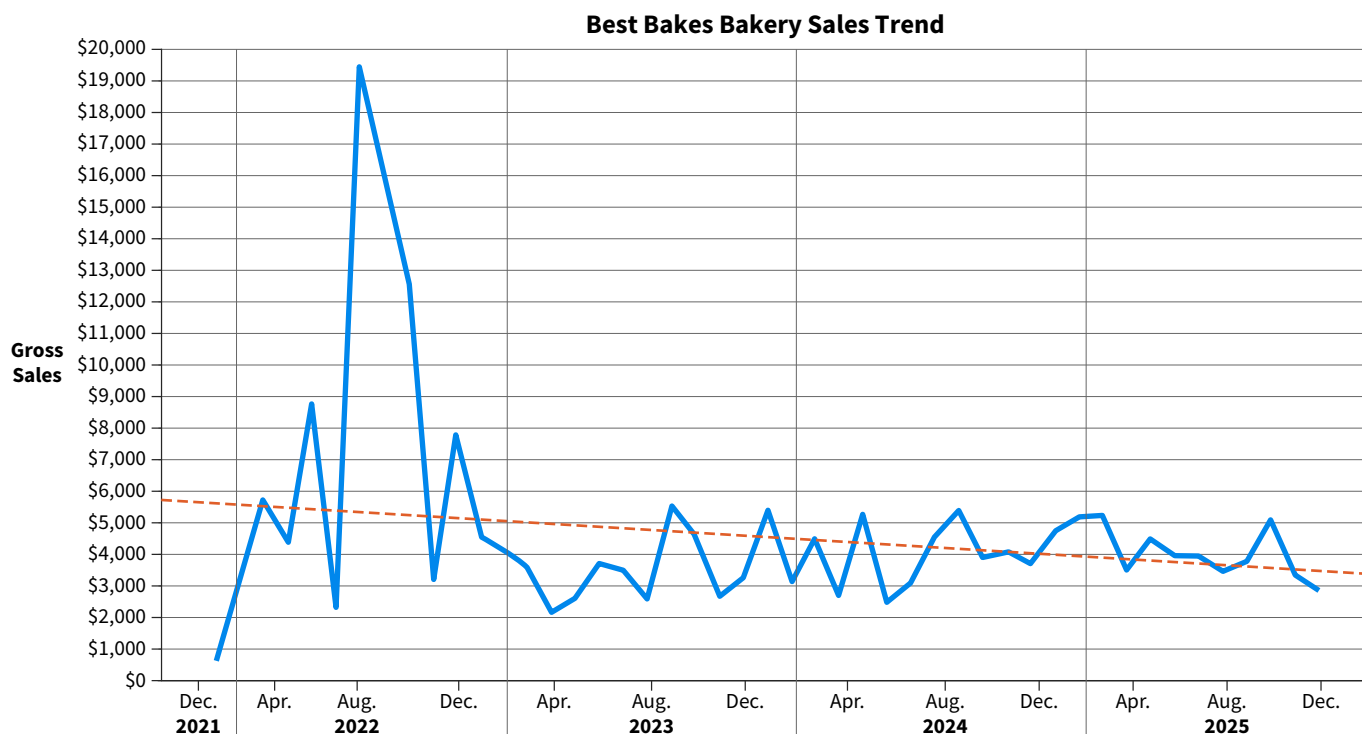
$$\text{Gross Sales} = \$74,739.90 - (\$1.56 \times \text{Month of Sale}).$$

Model statistics include:

$$R\text{-square} = 0.0487$$

$$p\text{-value} = 0.1406$$

The trendline equation was used to develop the following visualization. The month and year of sale is on the x-axis and gross sales are on the y-axis.



Examine the analysis in the provided data file and interpret the sales trend analysis.

1. What is the question/objective of the analysis?
2. Does the model and visualization provide adequate information to address it?
3. Are the model variables valid?

EX 8.20 (LO 5) Financial Accounting Managerial Accounting Interpret Regression Analysis You are asked to build a model to predict sales at U.S. Outdoor Adventures. Management believes that the region, discount percentage, and discount dollar amount are good predictors of sales. You used the data in the regression file to prepare the following regression analysis.

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.6260322				
R Square	0.3919164				
Adjusted R Square	0.391612				
Standard Error	486.1264				
Observations	9994				
ANOVA					
	df	SS	MS	F	Significance F
Regression	5	1521272575	304254515	1287.47445	0.00
Residual	9988	2360352937	236318.876		
Total	9993	3881625512			
	Coefficients	Standard Error	t Stat	P-value	
Intercept	221.75141	12.60104565	17.5978578	2.7291E-68	
Discount	-438.73427	24.64841598	-17.799694	8.4529E-70	
Discount \$	2.4154052	0.030141825	80.1346697	0.00	
Region West	0.1232541	15.12802917	0.0081474	0.99349955	
Region East	-17.727807	14.84929891	-1.1938481	0.23256576	
Region Central	18.590415	15.9032368	1.16897054	0.2424434	

1. Interpret the regression statistics.
2. Is the model is significant? Why or why not?
3. What are the coefficients of the model?
4. Do the variables used in the model make sense? Why or why not?
5. Identify other variables that may improve the model.
6. Based on this model, what is the predicted sale when the discount percentage is 20%, the sale is in the west region, and the discount dollar amount is 100?

Problems

PR 8.1 (LO 1) Data Financial Accounting Interpret Descriptive Analytics Management at U.S. Outdoor Adventures is evaluating their shipping policy to customers. Currently, the company offers free shipping, but they are considering charging customers for shipping. Management would like to better understand how shipping costs are changing and if there is a relationship between the type of shipping method and order priority.

Products are shipped either first class, same day, second class, or standard class. The method of shipping is decided by the customer but is usually related to the order priority. The priority categories are critical, low, high or medium.

The data analysis team provided the following descriptive analyses, which indicate that total shipping costs increased from 2022 to 2025. They also prepared a frequency distribution of the number of shipments by priority and shipping mode for 2022, 2023, 2024, and 2025 combined.

Shipping Cost Per Year

Year	Total Shipping Cost
2022	\$ 49,769.30
2023	\$ 50,198.57
2024	\$ 60,925.19
2025	\$ 77,280.73
Grand Total	\$ 238,173.79

Number of Shipments by Shipping Mode

Priority and Shipping Mode	Number of Shipments
Critical	783
First Class	340
Same Day	154
Second Class	289
High	3,069
First Class	703
Same Day	273
Second Class	744
Standard Class	1,349
Low	432
Standard Class	432
Medium	5,710
First Class	495
Same Day	116
Second Class	912
Standard Class	4,187
Grand Total	9,994

- Identify the following:
 - Who are the stakeholders ?
 - What was the purpose for analysis?
 - Are there alternative explanations or analyses?
 - Are there any risks that should be considered?
 - What knowledge did the preparer need and what knowledge does the interpreter need to understand the analysis?
 - How can you use your prior experience or apply this experience to future analyses through self-reflection?
- Are the data used in the analysis reasonable given the question/objective of the analysis?
- Is the analysis method reasonable given the question/objective of the analysis?
- Are the results of the analysis reasonable given what is known about the subject being analyzed?
- Are the implications of the analysis reasonable given what you know about the subject being analyzed?
- Does the analysis address the needs/concerns of the stakeholders?
- Prepare an additional descriptive analysis that will help U.S. Outdoor Adventures understand the costs associated with each mode of shipping. Interpret your analysis.

PR 8.2 (LO 1, 2, 3, 5) Data Financial Accounting Managerial Accounting Interpret Predictive Analyses All Care Hospital, which operates 2,000 hospitals in the United States, is preparing their operating budgets for 2026. Executive management would like to understand the factors driving total expenses. As a member of the data analytics team in the accounting department, you are asked to prepare and interpret data analyses to provide management with information about total hospital expenses.

The data analytics team provided descriptive analyses summarizing the number of hospitals in each region and total hospital expenses by region. In 2025, total expenses for All Care Hospitals was \$386.8 million. They also gave you the raw data file.

Region	Number of Hospitals
East North Central	262
East South Central	136
Mid Atlantic	168
Mountain	160
New England	100
Pacific	259
South Atlantic	252
West North Central	214
West South Central	449
Grand Total	2,000

Total Expenses by Region-2025	
Region	
East North Central	\$52,941,169
East South Central	\$19,037,967
Mid Atlantic	\$60,506,505
Mountain	\$26,438,517
New England	\$25,634,270
Pacific	\$71,210,066
South Atlantic	\$57,893,598
West North Central	\$26,015,206
West South Central	\$47,082,098

Total Expense

19M 71M

According to management, the following variables are predictors of total hospital expenses: admissions, census, outpatient visits, births, payroll expenses, personnel, and whether there was a birth or not. The analytics team prepared a regression model using those variables with the following output.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.982051941							
R Square	0.964426014							
Adjusted R Square	0.964301005							
Standard Error	61360.99522							
Observations	2000							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	7	2.03E+14	2.9E+13	7714.85352	0			
Residual	1992	7.5E+12	3.77E+09					
Total	1999	2.11E+14						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	-4191.07786	2275.171	-1.84209	0.06561006	-8653.042868	270.8871	-8653.04	270.8871
Admissions	6.013726131	0.387068	15.5366	1.8108E-51	5.254624717	6.772828	5.254625	6.772828
Census	-57.72122789	19.55346	-2.95197	0.00319445	-96.06861315	-19.3738	-96.0686	-19.3738
Outpatient Visits	-0.019727175	0.009985	-1.97573	0.04832264	-0.039308781	-0.00015	-0.03931	-0.00015
Births	-9.977260409	1.656594	-6.02276	2.0367E-09	-13.22609855	-6.72842	-13.2261	-6.72842
Payroll Expense	1.804388653	0.036122	49.95226	0.00000000	1.733547274	1.87523	1.733547	1.87523
Personnel	30.9359429	2.697418	11.46872	1.5657E-29	25.64588566	36.226	25.64589	36.226
Births or Not	-5932.874462	3329.682	-1.78181	0.07493183	-12462.8985	597.1496	-12462.9	597.1496

- Identify the following:
 - Who are the stakeholders?
 - What was the purpose for analysis?
 - Are there alternative explanations or analyses?
 - Are there any risks that should be considered?
 - What knowledge did the preparer need and what knowledge does the interpreter need to understand the analysis?
 - How can you use your prior experience or apply this experience to future analyses?
- Use the Summary Output for the following:
 - Interpret the regression statistics.
 - Is the model is significant? Why or why not?
 - Identify the coefficients of the model.
 - Do the variables used in the model make sense? Why or why not?
 - Identify other variables that may improve the model.
- Prepare a regression model that predicts total expenses using the variables: admissions, outpatient visits, births, and personnel.
 - Is this model better than the previous model?
 - Interpret the regression statistics and the coefficients of the model.

Professional Application Case: Ortho Inc.



Data Ortho Inc. is a medical device manufacturer based in San Diego, California. The company produces and sells custom-made knee replacement implants:

- It is a privately held company and files U.S GAAP based financial statements with its bank as part of its debt covenants.
- The board of directors is very involved and continues to emphasize the importance of well-designed and effective internal controls.
- Ortho is a decentralized organization, with 5 manufacturing and sales sites across the United States employing a total of 28 product managers and 29 sales managers. Each site operates independently and is responsible for procurement of raw materials, production, and sales of finished products in its own geographic region. Sales managers maintain their own customer relationships, while product managers monitor inventory levels and order raw materials as needed.
- Silicone and plaster are used to create the molds for all the company's knee replacements. The materials poured into molds vary by product—some are metal, some are plastic, and some are ceramic.

The company has invested significantly in the development and maintenance of its information system to collect and report complete and accurate data. Company management uses the data captured in the information system to create reports and support decision-making. The data dictionary includes the most relevant data used by management, accountants, auditors, and other professionals.

Data Dictionary

MATERIAL MASTER TABLE

Name	Primary Key	Definition
MaterialCategory	☐	Constant stating raw material
Material	☒	Material item identifier
MaterialDescription	☐	Description of material

Notes: The Material Master table contains information about different raw materials purchased by Ortho.

PRODUCT MASTER TABLE

Name	Primary Key	Definition
ProductCategory	☐	Constant stating knee implant
Product	☒	Product item identifier
ProductDescription	☐	Description of product

Notes: The Product Master table contains information about different products sold by Ortho.

TRANSACTIONS TABLE

Name	Primary Key	Definition
TransType	☒	Indicates the type of transaction (O for orders and S for sales)
TransNumber	☒	Identifier for orders and sales
SiteCode	☐	Transaction site (one of five locations)
Manager	☐	Employee associated with the transaction
EntityID	☐	Supplier/Customer associated with the transaction
Quantity	☐	Transaction quantity

Unit	☐	Unit of transaction (lb or kg for orders, ea for sales)
UnitPrice	☐	Price per unit
TransDate	☐	Sales/purchase order date
CompleteDate	☐	Shipping/receiving date (sales/orders)
Item	☐	The material purchased or product sold
CreatedBy	☐	The source of the data (e.g., daily batch)
Subtotal	☐	Transaction total before tax (UnitPrice * Quantity)
Tax	☐	Tax amount

Notes: The Transactions table contains a combination of purchase order, sales order, and sales return transactions

SITES TABLE

Name	Primary Key	Definition
SiteCode	☒	Identified for company sites (five state abbreviations)
SiteName	☐	Name of site (states)

Notes: The Sites table contains the name of each site, which is the name of the state where the site is located.

EMPLOYEE MASTER TABLE

Name	Primary Key	Description
EmployeeID	☒	Employee identifier
FirstName	☐	Employee first name
LastName	☐	Employee last name
SiteCode	☐	Company site identified (state)
Position	☐	Employee position
OrderApproval	☐	Employee order approval authority

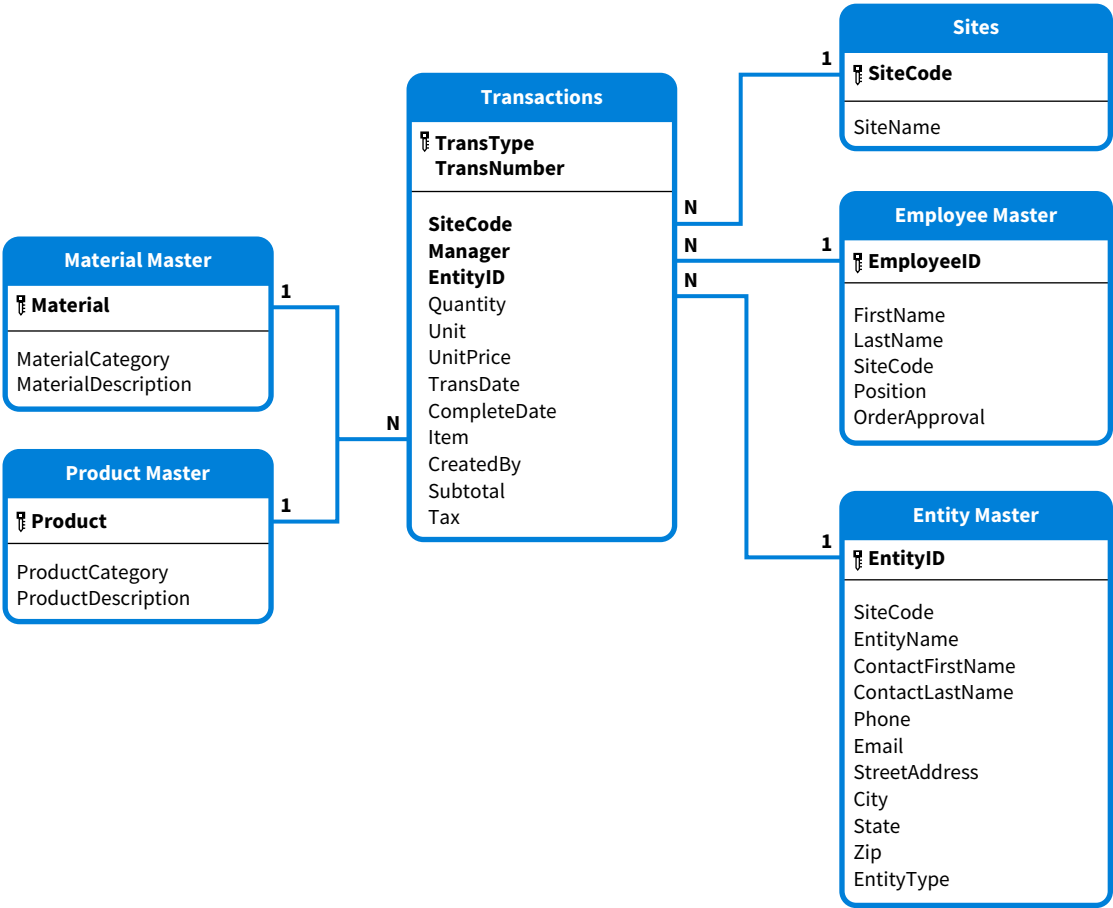
Notes: The Employee Master table contains information about employees associated with the sales order and purchase order transactions, e.g., sales and product managers.

ENTITY MASTER TABLE

Name	Primary Key	Description
EntityID	☒	Entity (supplier/customer/Ortho, Inc.) identifier
SiteCode	☐	Indicates which Ortho site the entity is associated with
EntityName	☐	Entity company name
ContactFirstName	☐	Entity contact first name
ContactLastName	☐	Entity contact last name
Phone	☐	Entity phone
Email	☐	Entity email
StreetAddress	☐	Entity street address
City	☐	Entity city
State	☐	Entity state
Zip	☐	Entity zip code
EntityType	☐	Entity type ("S" for customer and "O" for supplier)

The entity-relationship diagram (ERD) shows the relationships among the tables in the information system.

Entity-Relationship Diagram (ERD)



(Notes: The ERD indicates which tables to join and which columns to use in the joins.)

The comparative income statement shows income for 2024 and 2025.

Ortho Inc. Comparative Income Statement

Comparative Income Statement \$ in Millions		
	2024 (audited)	2025 (unaudited)
Net Sales	\$913	\$854
Cost of Sales	579	542
Gross Profit	334	312
Research and development	57	59
Selling, general and administrative	191	196
Recall charges, net of insurance proceeds	5	5
Intangible asset amortization	3	2
Total operating expenses	256	262
Operating Income	78	50
Other income (expenses), Net	1	2
Earnings before income taxes	79	52
Income taxes	11	8
Net earnings	\$ 68	\$ 44

PAC 8.1 Auditing: Interpret Descriptive and Diagnostic Analytics

Data Auditing You are on the external audit team auditing Ortho's financial statements as of and for the year ended December 31, 2025:

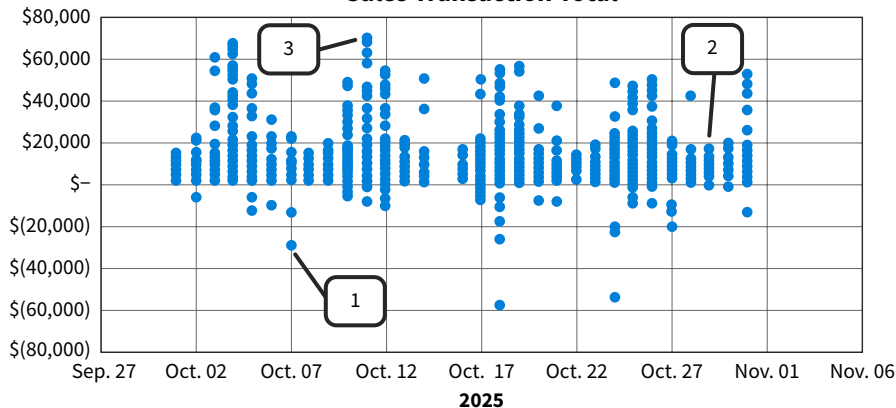
- Note that the amounts presented in the comparative income statement for 2024 are audited, whereas the amounts presented for 2025 are unaudited.
- For the year ended December 31, 2024, your audit firm issued an unqualified opinion on the financial statements and noted that the company's internal controls over financial reporting were appropriately designed and operating effectively.

You have been asked to consider risks of material misstatement related to net sales. Specifically, your audit senior has asked you to interpret the descriptive analytics provided by your firm's data analysis team. The Tableau output is provided.

Sales Frequency Distribution

Prod ProductDescription	TR TransDate			
	Distinct Count of TR Subtotal		TR Subtotal	
	2024	2025	2024	2025
Ceramic-on-Ceramic Knee	4,378	4,196	113,844,112	116,040,741
Metal and Crosslinked Polyethylene Knee	7,217	7,208	145,197,885	153,572,138
Metal and Plastic Knee	9,567	9,714	103,546,449	107,545,175
Metal-on-Metal Knee (Cobalt chrome)	20,457	20,331	185,198,302	198,397,781
Metal-on-Metal Knee (Stainless steel)	4,153	4,306	106,320,616	116,722,920
Metal-on-Metal Knee (Titanium)	16,526	16,760	199,398,247	220,828,360
Grand Total	60,102	60,424	853,505,611	913,107,114

1. Consider the validity of the Tableau output provided by your information technology team. Is the sales frequency distribution table analysis valid and reliable? Why or why not?
2. Compare the sales mix from 2025 to 2024. How does sales mix contribute to consideration of the risk of material misstatement?
3. During inquiry with the vice president of sales, your engagement team learned that the largest customer purchases only metal-on-metal knee products, and that these purchases account for approximately 35% of all sales of these products. This customer has recently requested significant modifications to their sales order contract and credit terms because of possible bankruptcy concerns. How might this information, combined with the sales frequency distribution table, contribute to the risk of material misstatement?
4. The technology team has provided a scatterplot of sales for the metal-on-metal knee (cobalt chrome) product for the month of October. The technology team has identified certain points on the diagram. Explain whether each of the identified points on the diagram would need further investigation, or if they should be included in normal substantive testing for sales.

Sales Transaction Total

5. Use the Ortho, Inc data sets to create additional descriptive analytics. Consider creating a cross tabulation analysis of sales by product description and/or sales by site code. These descriptive statistics provide insight into the company's net sales for the year under audit. Describe your analysis, and interpret your results.

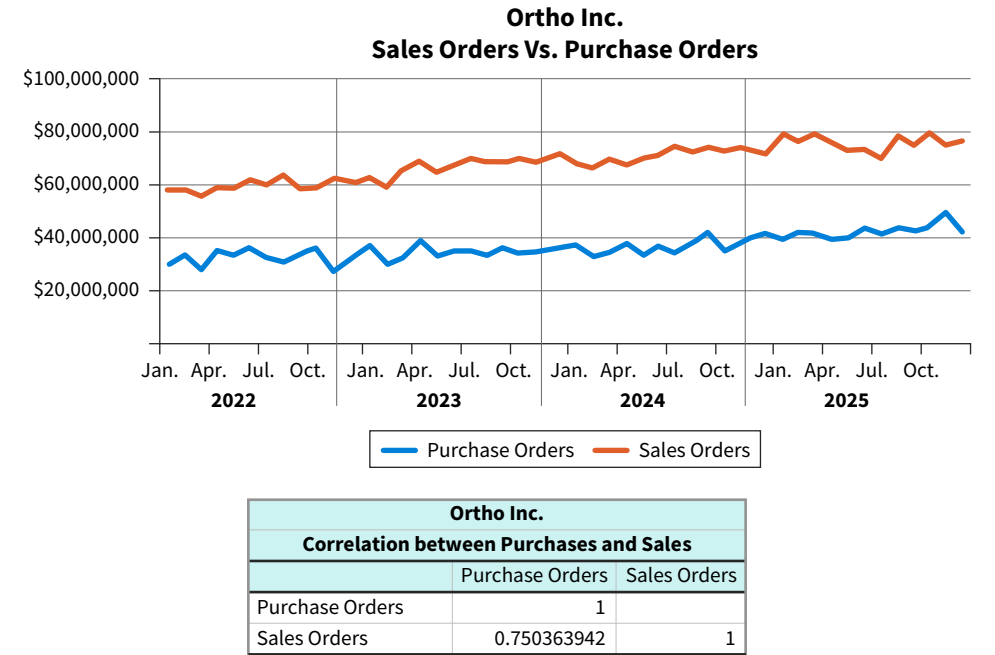
PAC 8.2
Managerial Accounting: Interpret Diagnostic Analytics

Data

Managerial Accounting

You have been tasked with understanding the relationship between sales orders and purchase orders to acquire raw materials. Currently, production sites are individually responsible for all purchases. The benefit of this decentralized organization is that the sites can quickly respond to changes in production demand. The downside is that sourcing for purchases is not coordinated between the sites. There is a concern that some sites' purchasing departments have been buying more raw materials than needed.

Your manager has asked you to analyze purchases compared to sales. You prepared the following analyses.



1. How do these analyses help evaluate if the sites are purchasing more raw materials than they need?

2. List some additional analyses that would be useful.

3. Create additional analyses and interpret your results.

PAC 8.3
Financial Accounting: Interpret Descriptive Analytics

Data

Financial Accounting

The Controller at Ortho Inc. has asked you to understand the requirements of ASC 606 Revenue Recognition regarding the disclosure of sales data by geographic or product categories. You have prepared the cross tabulation analysis for 2025 and 2024 as a starting point for preparing the disclosure.

Site	2025	2024	Difference
CA	172,934,662.11	174,888,045.18	-1,953,383.07
FL	183,493,309.15	164,530,136.81	18,963,172.34
IL	175,041,416.59	174,872,628.25	168,788.34
NY	193,364,104.19	163,660,289.73	29,703,814.46
TX	188,249,027.51	175,554,510.70	12,694,516.81
Total	913,082,519.55	853,505,610.67	59,576,908.88

1. Who are the key stakeholders of the analysis?

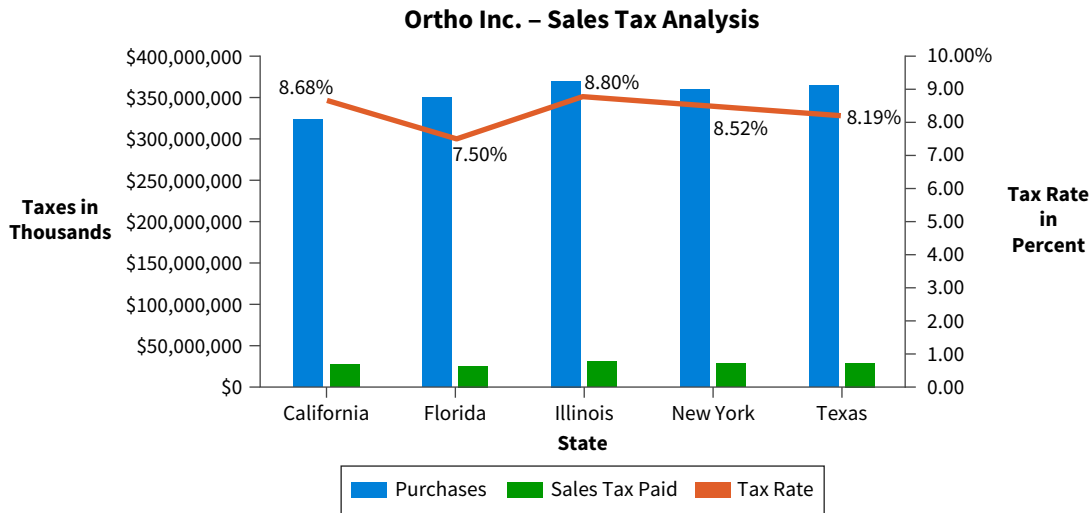
2. What knowledge do you need to provide a valid interpretation of the descriptive analysis?

3. Is this analysis sufficient for segment reporting or will you need additional analyses?

4. Create additional analyses of results by year.

PAC 8.4 Tax Accounting: Interpret Descriptive and Diagnostic Analytics

Data Tax Accounting Ortho's management would like to identify potential opportunities for cost savings. One aspect of purchasing costs that management has not yet considered is sales tax paid on purchases. Management has requested an analysis of sales tax paid by site. As a staff tax accountant for Ortho, you have prepared the following trend analysis of sales tax paid by location.



1. Prepare an interpretation of the analysis.
2. Is this analysis sufficient or will you need additional analyses?
3. Create additional analyses of results by year.

Le Grind Continuing Case: Verify and Interpret the Gross Profit Analysis

Data Visit Wiley's online learning platform for additional questions, data, and more details about the continuing case.

